



Section 4
Health effects

Guideline No. 497

Defined Approaches on Skin Sensitisation

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OECD GUIDELINES FOR THE TESTING OF CHEMICALS

Guideline No. 497: Defined Approaches on Skin Sensitisation

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Section 1-Introduction

1.1. General Introduction

1. A skin sensitiser refers to a substance that will lead to an allergic response following repeated skin contact as defined by the United Nations Globally Harmonized System of Classification and Labelling of Chemicals (UN GHS) (1). There is general agreement on the key biological events underlying skin sensitisation. The current knowledge of the chemical and biological mechanisms associated with skin sensitisation initiated by covalent binding to proteins has been summarised as an Adverse Outcome Pathway (AOP) (2) that begins with a molecular initiating event, leading to intermediate key events (KEs), and terminating with the adverse effect, allergic contact dermatitis.

2. The skin sensitisation AOP focuses on chemicals that react with amino acid residues (*i.e.* cysteine or lysine) such as organic chemicals. In this instance, the molecular initiating event (*i.e.* the first KE), is the covalent binding of electrophilic substances to nucleophilic centres in skin proteins. The second KE in this AOP takes place in the keratinocytes and includes inflammatory responses as well as changes in gene expression associated with specific cell signalling pathways such as the antioxidant/electrophile response element (ARE)-dependent pathways. The third KE is the activation of dendritic cells, typically assessed by expression of specific cell surface markers, chemokines and cytokines. The fourth key event is T-cell proliferation, and the adverse outcome is presentation of allergic contact dermatitis.

3. The assessment of skin sensitisation has typically involved the use of laboratory animals. The classical methods that use guinea pigs, the Guinea Pig Maximisation Test (GPMT) and the Buehler Test (OECD TG 406) (3) assess both the induction and elicitation phases of skin sensitisation. The murine tests, such as the local lymph node assay (LLNA) (OECD TG 429) (4) and its three non-radioactive modifications — LLNA: DA (OECD TG 442A) (5), LLNA: BrdU-ELISA, and BrdU-FCM (OECD TG 442B) (6) — all assess the induction phase exclusively and have gained acceptance, since they provide an advantage over the guinea pig tests in terms of animal welfare together with an objective measurement of the induction phase of skin sensitisation.

4. Mechanistically-based *in chemico* and *in vitro* test methods (OECD TG 442C, 442D, 442E) (7, 8, 9) addressing the first three KEs of the skin sensitisation AOP can be used to evaluate the skin sensitisation hazard potential of chemicals. With the exception of the kinetic Direct Peptide Reactivity Assay (k-DPRA) (TG442C), none of these test methods are considered sufficient stand-alone replacements of animal data to conclude on the skin sensitisation potential of chemicals or to provide information for potency sub-categorisation according to the UN GHS (sub-categories 1A and 1B). However, data generated with these *in chemico* and *in vitro* methods addressing multiple KEs of the skin sensitisation AOP can be used together, as well as with information sources such as *in silico* and read-across predictions from chemical analogues, within integrated approaches to testing and assessment (IATA) or defined approaches (DAs). Results from the individual information sources can only be used in DAs if the substances fall within the applicability domains of the methods, which are described in the “Initial Considerations, Applicability and Limitations” sections of the respective methods (TG 442C, Appendices I-II; TG 442D, Appendices IA-IC; TG 442E Annexes 1-4) (7, 8, 9), and in Appendices I-V of this guideline for the *in silico* information sources.

5. Results from multiple information sources can be used together in DAs to achieve an equivalent or better predictive capacity than that of the animal tests to predict responses in humans. A DA consists of a fixed data interpretation procedure (DIP) (e.g. a mathematical model, a rule-based approach) applied to data (e.g. *in silico* predictions, *in chemico*, *in vitro* data) generated with a defined set of information sources to derive a prediction without the need for expert judgment. Individual DAs for skin sensitisation and their respective information sources were originally described in Guidance Document 256, Annex I/II (10) and a preliminary assessment was published in Kleinstreuer et al (11). The DAs use method combinations intended to overcome some of the limitations of the individual methods to provide increased confidence in the overall result obtained. The ultimate goal of DAs is to provide information that is equivalent to that provided by animal studies, *i.e.* information that can be used for hazard identification, potency (sub)-categorisation, and/or a quantitative measure of sensitisation potency, used to derive a point of departure (PoD).
6. Testing laboratories should consider all relevant available information on the test chemical prior to conducting the studies as directed by a DA. All available information could include, for example, the identity and chemical structure of the test chemical and its physicochemical properties. Such information should be considered in order to determine whether the individual OECD test guideline methods and any *in silico* information source used under a specific DA are applicable for the test chemical.
7. When performing a hazard evaluation, potency (sub)-categorisation, and/or PoD prediction based on the output from an *in vivo* (e.g. LLNA) test, an *in chemico* test, an *in vitro* test, an *in silico* approach, a DA, or any combination thereof, the same principles always apply. That is, all available information relevant to the chemical in question as well as toxicological data on structurally related test chemicals, if available, should be taken into consideration.
8. This Guideline was developed with the input of an OECD Expert Group on Defined Approaches for Skin Sensitisation (EG DASS) comprised of scientific experts from regulatory agencies, validation bodies, non-governmental organisations, and industry.
9. Multiple rule-based DAs are included in this Guideline and are described in the context of their primary intended regulatory purpose: hazard identification, *i.e.* discrimination between skin sensitisers and non-sensitisers (Part I), potency sub-categorisation (Part II), or derivation of a point of departure (Part III)¹. The DA included in Part II is also suitable for hazard identification (Part I). The DA(s) included in Part III cannot be used in Part I or Part II unless mentioned specifically for their use. The evaluation and review of the DAs are described in detail in the *Supporting document to the Guideline (GL) on Defined Approaches (DAs) for Skin Sensitisation* (12).
10. A comprehensive dataset of 196 chemicals with DA predictions, data on individual information sources, highly curated LLNA and Human Patch Predictive Test (HPPT) data, and physicochemical properties, was compiled and is attached as **Annex 2** to the *Supporting document to the Guideline (GL) on Defined Approaches (DAs) for Skin Sensitisation* (12). Information on the curation of this data is detailed in **Annexes 3 and 4** of the *Supporting document to the Guideline (GL) on Defined Approaches (DAs) for Skin Sensitisation* (12). Out of the 196 chemicals, 168 chemicals have classifications based on LLNA data and 66 chemicals have classifications based on human data, which were all agreed upon by the EG DASS and used to evaluate the performance of the DAs when using the initially included (see paragraph 11) test methods and *in silico* models. Due to the availability of data, this dataset contains mainly cosmetic ingredients but also other types of chemicals that are used across sectors such as preservatives, dyes, or food ingredients. The dataset is chemically diverse as shown by the range of physicochemical properties covered by these chemicals: it contains small and large molecules (molecular weight ranges from 30 to

¹ Point of departure – The point on a dose response curve established from experimental data used to derive a safe level

512 g/mol), hydrophilic and hydrophobic substances (Log P ranges from -3.9 to 9.4), liquids and solids (melting point ranges from -122 to 253 °C), volatile and non-volatile substances (boiling point ranges from -19 to 445 °C). Further details on the chemical space characterization of the reference database are available in **Section 4** of the *Supporting document to the Guideline (GL) on Defined Approaches (DAs) for Skin Sensitisation* (12).

11. This Guideline, when initially adopted, included DAs developed to be performed with a specific set of methods, *i.e.* the Direct Peptide Reactivity Assay (DPRA), KeratinoSens™, and the Human Cell Line Activation Test (h-CLAT), for the “2 out of 3” approach (2o3), and DPRA, h-CLAT, and either Derek Nexus or OECD quantitative structure-activity relationship ((Q)SAR) Toolbox for the integrated testing strategy (ITS). These DAs are referred as ‘classic DAs’ together or ‘classic 2o3 DA’ and ‘classic ITS DA’ throughout this Guideline. In the course of evaluating and developing the first DAs and specific information sources, a number of analyses were conducted by the EG DASS members. These include the drafting of an evaluation framework for each DA, building, curating, and characterising chemical reference lists of human and murine data, and developing decision trees and other tools to incorporate *in silico* information sources into the relevant DAs and guideline. The group also identified sources of uncertainty and potential limitations of the DAs and information sources, as well as borderline test results, and assessed the impact of these on the performance of the DAs. These analyses are specific to the creation of the original DASS guideline and information sources published in 2021 and can be found in the main document and Annexes 1 & 3-8 of the *Supporting document to the Guideline (GL) on Defined Approaches (DAs) for Skin Sensitisation* (12).

12. In 2025, this guideline was updated to (i) include similar methods² to the methods (see paragraph 11) used for the initial adoption of this guideline and (ii) to include part III on Defined Approaches for derivation of PoD. Based on the 2025 revision, Annex 2 of the supporting document has been updated with new performance information relevant to the new information sources and DAs that have been added to the guideline.

13. Since the original guideline adoption, other *in chemico* (included in TG 442C) and *in vitro* (included in TGs 442D/E) methods have been proposed and inserted into this Guideline as alternative information sources in the DAs (see **Table 1.1** and subsequent sections of this document for the list of methods permitted for use in each DA). The performances of the DAs using the alternative information sources have been evaluated with available datasets for each permutation and can be found in **Tables 1.1-1.3** below and in **Annexes 2** and **9** of the *Supporting document to the Guideline (GL) on Defined Approaches (DAs) for Skin Sensitisation* (12).

1.2 DAs included in the Guideline

1.2.1 DAs in Part I and Part II.

14. The DAs currently described in **Part I** and **Part II** of this guideline are:

- The 2o3 defined approach to skin sensitisation hazard identification based on *in chemico* (KE1) and *in vitro* (KE2/KE3) data (13, 14). See **Part I**.

² Similar methods in this guideline refer to methods that target a same key event (KE) and/or that are based on a similar technology.

- The integrated testing strategy (ITS) for UN GHS potency categorisation based on *in chemico* (KE1) and *in vitro* (KE3) data, and *in silico* predictions (15), with a DIP developed with EG DASS input. See **Part II**.

15. The DAs described in this guideline are based on the use of validated OECD test methods for which transferability, within- and between-laboratory reproducibility have been characterised in the validation phase (OECD 442C/D/E, (7, 8, 9)).

16. The ITS DAs make use of well-defined *in silico* information sources. Information on each model, including its description, applicability domain, and protocol to generate predictions is available in Appendices I-II. Additionally, key information related to the OECD Principles for the Validation, for Regulatory Purposes, of (Q)SAR Models (16), is described in individual QSAR Model Reporting Format (QMRF) documents (17).

17. All DAs described in Parts I and II of this guideline can be used to address countries' requirements for discriminating sensitisers (*i.e.* UN GHS Category 1) and non-sensitisers (UN GHS Not Classified (NC)), though they do so with different sensitivities and specificities (detailed in the respective descriptions of each DA and in **Annexes 2** and **9** of the *Supporting document to the Guideline (GL) on Defined Approaches (DAs) for Skin Sensitisation* (12)).

18. The ITS DA can also be used to discriminate chemicals into three UN GHS potency categories (Category 1A = strong sensitisers; Category 1B = other sensitisers), and NC.

19. The known limitations and applicability domains of the individual information sources were used to design workflows for assigning confidence to each of the predictions produced by the DAs described in this guideline. In order to have a high confidence prediction, the underlying data must meet criteria in the respective *in chemico/in vitro* TGs (see TG 442C, Appendices I-II; TG 442D, Appendices IA-IC; TG 442E Annexes 1-4 (7, 8, 9)); for the ITS DA, *in silico* predictions must fall within the applicability domain of the model used (see Appendices I-II). DA predictions with high confidence for hazard identification and/or potency categorisation are considered conclusive. DA predictions with low confidence are considered inconclusive for hazard identification and/or potency categorisation (see **Sections 2.1.4** and **3.1.4** for further information). These inconclusive predictions may nevertheless be considered in a weight-of-evidence approach and/or within the context of an IATA together with other information sources (*e.g.* demonstration of exposure to the test system, existing *in vivo* data, clinical data, read-across, other *in chemico/in vitro/in silico* data, etc.).

20. The performance of the classic DAs in this guideline (see paragraph 11) were evaluated using a curated dataset of up to 168 test chemicals with classifications based on LLNA and up to 66 chemicals with classifications based on human data and corresponding information source data, where available. For additional details see **Annex 3** of the *Supporting document to the Guideline (GL) on Defined Approaches (DAs) for Skin Sensitisation* (12). The performances of the DAs using the alternative information sources³ have been evaluated against both LLNA and human-based reference data, according to available datasets for each permutation, and can be found in Table 1.1 below and in **Annexes 2** and **9** of the *Supporting document to the Guideline (GL) on Defined Approaches (DAs) for Skin Sensitisation* (12).

21. The performance range of the 2o3 and ITS DAs (high confidence predictions only) against the LLNA-based reference data for predicting skin sensitisation hazard (2o3 and ITS DAs) and potency (ITS DA only) is summarized in **Table 1.1**. Note that specificity measures are more uncertain than sensitivities due to lower number of negative reference chemicals. In addition, **Table 2** and **Table 1.3** show the Balanced Accuracy (BA) for hazard, and Accuracy (Acc) for potency, of all 2o3 and ITS DA permutations

³ Alternative Information Sources: Information sources based on similar methods to the specific set of methods used for the initial adoption of this guideline for Part I and Part II

against LLNA-based reference data and human-based reference data, respectively. For the ITS DA, there were no strong sensitisers (1A) that were incorrectly predicted as being a non-sensitiser (NC) or vice versa. Detailed performance statistics for each permutation of the ITS DA are reported in **Annex 2** and **9** the *Supporting document to the Guideline (GL) on Defined Approaches (DAs) for Skin Sensitisation* (12).

22. The performance range of the 2o3 and ITS DAs (high confidence predictions only) against the human-based reference data for predicting skin sensitisation hazard (2o3 and ITS DAs) and potency (ITS DA only). Two ITS permutations (**Table 1.3**) had lower accuracy than the LLNA when compared to human-based reference data which was due to several underpredictions of 1A sensitisers as 1B sensitisers. These underpredictions may be attributed to several considerations, including human-based sub-categorizations derived from human data where the predicted sensitising dose was close to the threshold between GHS 1A and GHS 1B sub-categorizations (termed GHS 1A- in **Annex 4** of the *Supporting document to the OECD Guideline 497 on Defined Approaches for Skin Sensitisation* for more information on how this was derived). This lower accuracy is specific to these ITS DA permutations only and should be taken into consideration when selecting specific DAs to use. The underprediction cases are discussed further in **Annex 9** of the *Supporting Document to the Guideline (GL) on Defined Approaches (DAs) for Skin Sensitisation* (12). Detailed performance statistics for each permutation of the 2o3 and ITS DAs are reported in **Annexes 2** and **9** of the *Supporting document to the Guideline (GL) on Defined Approaches (DAs) for Skin Sensitisation* (12).

23. The performance of the LLNA against the human-based reference data for predicting skin sensitisation hazard (2o3 and ITS DAs) and potency (ITS DA only). Note that the specificity measure is more uncertain than the sensitivity due to a lower number of negative reference chemicals. Detailed performance statistics are reported in **Annex 2** of the *Supporting Document to the Guideline (GL) on Defined Approaches (DAs) for Skin Sensitisation* (12).

Table 1.1. Summary of the DAs Included in Part I/II of this Guideline

DA/ Method	Information Sources (one each per KE and/pr <i>in silico</i> , where applicable)	Capability	Hazard Performance vs. LLNA	Hazard Performance vs. Human	Potency Performance vs. LLNA	Potency Performance vs. Human
2o3 DA	KE1: ADRA, DPRA KE2: KeratinoSens™, LuSens, EpiSensA KE3: GARD®skin, h-CLAT, IL-8 Luc, U-SENS™	Hazard	80-95% BA, 82-97% Sens, 72-100% Spec	72-97% BA, 84-97% Sens, 50-100% Spec	-	-
ITS DA	KE1: ADRA, DPRA KE3: GARD®skin, h-CLAT, U-SENS™ <i>In silico</i> : Derek Nexus, OECD QSAR Toolbox	Haza rd, Pote ncy	78-88% BA, 91-94% Sens, 65-82% Spec	67-76% BA, 87-96% Sens, 44-57% Spec	66-80% Accuracy 4-17% overpredi cted 13-18% underpredi cted	54-75% Accuracy 13-18% overpredict ed 13-31% underpredi cted
LLNA (provided for compariso n)	<i>in vivo</i>	Haza rd, Pote ncy	-	58% BA, 94% Sens, 22% Spec	-	60% Accuracy 19% overpredict ed 21% underpredi cted

Note: The performance ranges are illustrated here, when evaluated against the maximal common dataset⁴. For individual permutations of each DA, please refer to **Annexes 2 and 9** of the *Supporting document to the Guideline (GL) on Defined Approaches (DAs) for Skin Sensitisation (12)*. For hazard performance, sensitivity (Sens) is the true positive rate, specificity (Spec) is the true negative rate, and balanced accuracy (BA) is the average of sensitivity and specificity. Due to the imbalanced nature of the reference data, the measures of specificity are more uncertain than the measures of sensitivity. For potency performance, accuracy reflects the overall correct classification rate for all UN GHS sub-categories (Category 1A = strong sensitisers; Category 1B = other sensitisers, and Not Classified (NC). Overprediction refers to chemicals misclassified as more potent, i.e., a NC predicted to be a 1A/1B or a 1B sensitiser predicted to be a 1A sensitiser. Underprediction refers to chemicals misclassified as less potent, i.e., a 1A sensitiser predicted to be a 1B/NC, a 1B sensitiser predicted to be NC. Due to the imbalanced nature of the reference data, the measures of Specificity, and consequently also Balanced Accuracy, are more uncertain for classes that contain a low number of chemicals, e.g. NC chemicals. This is especially true for the human reference data set. Statistics reflect conclusive DA predictions only. This represents the information sources evaluated for inclusion under the initial adoption and 2025 update of this guideline.

Model Abbreviations: ADRA: Amino acid Derivative Reactivity Assay; DPRA: Direct Peptide Reactivity Assay; EpiSensA: Epidermal Sensitization Assay; GARD: Genomic Allergen Rapid Detection; h-CLAT: Human Cell Line Activation Test; IL-8 Luc: Interleukin-8 Reporter Gene Assay; U-SENS: U-937 cell line activation test.

24. **Table 1.1** provides an overview of the DAs included in this Guideline, their information sources used, whether they provide hazard and/or potency prediction, and summarises their performance range against the LLNA and human-based reference data. The LLNA performance against human-based reference data is included in **Table 1** as a basis for comparison. More details are provided in Part I and

⁴ Datasets which use all available data points for a specific permutation of the 2o3 or ITS DA (see **Annex 9** of the *Supporting document to the Guideline (GL) on Defined Approaches (DAs) for Skin Sensitisation (12)*).

Part II of this Guideline, as well as in the *Supporting document to the Guideline (GL) on Defined Approaches (DAs) for Skin Sensitisation* (12).

Table 1.2. List of the 203 DA Permutations Included in this Guideline and Their Balanced Accuracy (hazard) Against LLNA- and Human-based Reference Data

203 DA Permutation	BA % (LLNA)	BA % (Human)
ADRA, EpiSensA, GARDskin	93.2	80.3
ADRA, EpiSensA, IL-8	87.9	76.7
ADRA, EpiSensA, U-SENS	83.9	86.1
ADRA, EpiSensA, h-CLAT	88.3	82.8
ADRA, KeratinoSens, GARDskin	88.4	80.5
ADRA, KeratinoSens, IL-8	88.3	81.3
ADRA, KeratinoSens, U-SENS	80.1	85.9
ADRA, KeratinoSens, h-CLAT	87.3	81.9
ADRA, LuSens, GARDskin	94.9	78.0
ADRA, LuSens, IL-8	90.8	85.6
ADRA, LuSens, U-SENS	83.6	92.1
ADRA, LuSens, h-CLAT	92.3	84.3
DPRA, EpiSensA, GARDskin	92.7	78.5
DPRA, EpiSensA, IL-8	88.6	71.9
DPRA, EpiSensA, U-SENS	83.8	96.2
DPRA, EpiSensA, h-CLAT	89.4	81.9
DPRA, KeratinoSens, GARDskin	85.3	87.1
DPRA, KeratinoSens, IL-8	85.3	89.0
DPRA, KeratinoSens, U-SENS	80.9	86.9
DPRA, KeratinoSens, h-CLAT*	83.5	88.4
DPRA, LuSens, GARDskin	94.6	86.4
DPRA, LuSens, IL-8	91.4	97.0
DPRA, LuSens, U-SENS	83.3	94.7
DPRA, LuSens, h-CLAT	93.3	97.4
LLNA		57.9

Note: The accuracy of each 203 DA permutation is illustrated here when evaluated against the maximal common dataset⁵. For more detailed performance statistics of each DA permutation, please refer to **Annexes 2 and 9** of the *Supporting document to the Guideline (GL) on Defined Approaches (DAs) for Skin Sensitisation* (12). For hazard performance, balanced accuracy (BA) is the average of sensitivity and specificity. This represents the information sources evaluated for inclusion under the update of this guideline in 2025. * refers to the classic 203 DA, Model Abbreviations: ADRA: Amino acid Derivative Reactivity Assay; DPRA: Direct Peptide Reactivity Assay; EpiSensA; Epidermal Sensitization Assay; GARD: Genomic Allergen Rapid Detection; h-CLAT: Human Cell Line Activation Test; IL-8 Luc: Interleukin-8 Reporter Gene Assay; U-SENS: U-937 cell line activation test.

⁵ Datasets which use all available data points for a specific permutation of the 203 or ITS DA (see **Annex 9** of the *Supporting document to the Guideline (GL) on Defined Approaches (DAs) for Skin Sensitisation* (12)).

Table 1.3. List of the ITS DA Permutations Included in this Guideline and Their Balanced Accuracy (hazard) and Accuracy (potency) Against LLNA- and Human-based Reference Data

ITS DA Permutation	Hazard		Potency	
	BA (LLNA) %	BA (Human) %	Acc % (LLNA)	Acc % (Human)
ADRA, GARDskin, Derek	82.2	70.7	77.3	62.2
ADRA, GARDskin, OECD.TB	84.9	70.5	79.2	60.5
ADRA, U-SENS, Derek	81.5	67.6	70.4	55.2
ADRA, U-SENS, OECD.TB	82.3	70.0	70.3	53.8
ADRA, h-CLAT, Derek	82.8	67.7	73.0	62.7
ADRA, h-CLAT, OECD.TB	82.1	71.2	72.8	64.9
DPRA, GARDskin, Derek	81.5	76.5	78.7	75.0
DPRA, GARDskin, OECD.TB	87.8	76.4	79.7	73.7
DPRA, U-SENS, Derek	79.8	69.4	67.6	66.1
DPRA, U-SENS, OECD.TB	78.0	68.4	66.2	63.6
DPRA, h-CLAT, Derek*	80.7	68.6	70.2	72.7
DPRA, h-CLAT, OECD.TB*	79.8	69.4	70.6	71.4
LLNA		57.9		59.6

Note: The accuracy of each ITS DA permutation is illustrated here when evaluated against the maximal common dataset⁶. For more detailed performance statistics of each DA permutation, please refer to **Annexes 2 and 9** of the *Supporting document to the Guideline (GL) on Defined Approaches (DAs) for Skin Sensitisation (12)*. For hazard performance, balanced accuracy (BA) is the average of sensitivity and specificity. For potency performance, accuracy (Acc) reflects the overall correct classification rate for all UN GHS sub-categories (Category 1A = strong sensitisers; Category 1B = other sensitisers, and Not Classified (NC). This represents the information sources evaluated for inclusion under the update of this guideline in 2025. * refers to the classic ITS DAs.

Model Abbreviations: ADRA: Amino acid Derivative Reactivity Assay; DPRA: Direct Peptide Reactivity Assay; EpiSensA; Epidermal Sensitization Assay; GARD: Genomic Allergen Rapid Detection; h-CLAT: Human Cell Line Activation Test; IL-8 Luc: Interleukin-8 Reporter Gene Assay; U-SENS: U-937 cell line activation test.

1.2.2. DAs in Part III

25. The DAs currently described in **Part III** of this guideline are:

- The Skin Allergy Risk Assessment – Integrated Chemical Environment (SARA-ICE) defined approach which is a Bayesian Statistical model for the derivation of a quantitative point of departure based on *in Chemico* (KE1), *in vitro* (KE2/KE3), and/or (existing) *in vivo* (KE4/AO) data (4-9). See **Part III**.
- The “Regression-based” Defined Approach (DA) derives a point of departure for skin sensitisation based on quantitative (dose-response and kinetic) data from *in vitro* and *in chemico* assays addressing KE1, KE2, and KE3 of the AOP. See **Part III**.

26. The SARA-ICE DA in **part III** allows for the use of existing *in vivo* data from the LLNA or from human predictive patch tests as available; no new *in vivo* data should be generated for use in this DA.

⁶ Datasets which use all available data points for a specific permutation of the 2o3 or ITS DA (see **Annex 9** of the *Supporting document to the Guideline (GL) on Defined Approaches (DAs) for Skin Sensitisation (12)*).

27. SARA-ICE in **Part III** is a Bayesian statistical model which estimates a probabilistic distribution from which a quantitative point of departure for skin sensitisation risk assessment can be derived. SARA-ICE is freely available for download within a graphical user interface ([LINK TO SARA-ICE APP⁷](#))

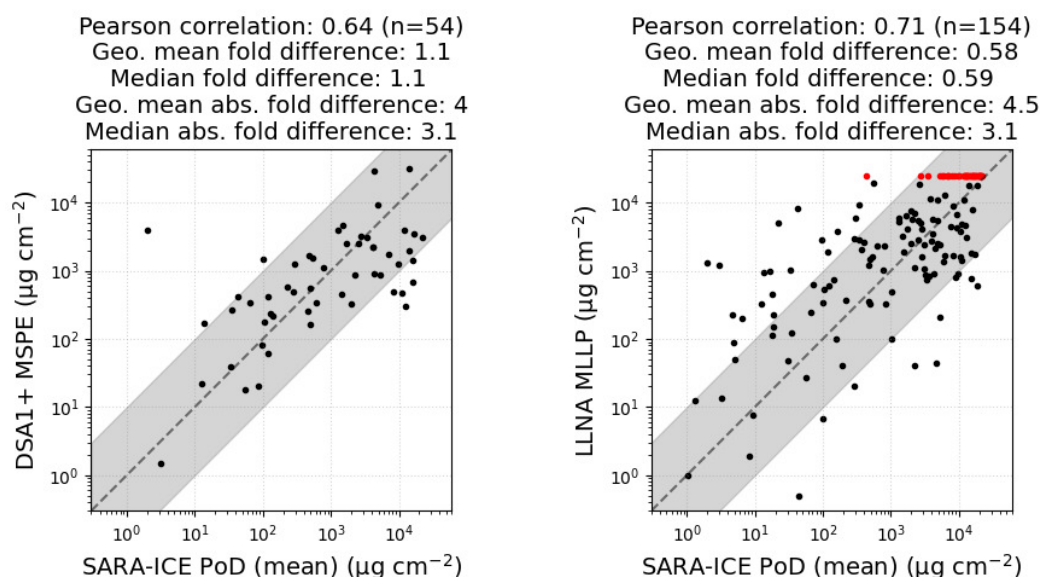
28. The SARA-ICE DA derives a point of departure for risk assessment. The model estimates the dose at which skin sensitisation is likely to occur in 1% of an eligible human population (estimated dose, ED₀₁), reported as a distribution with associated confidence intervals. An upper bound is imposed on ED₀₁ values, at a threshold of 60,000 µg cm⁻². This reflects the bound for a GHS classification of NC and an implicit assumption that the chemical under consideration for PoD based risk assessment is a sensitiser at some exposure level. The PoD is therefore defined as the best estimate (geometric mean) of the distribution of the ED₀₁ truncated at 60,000 µg cm⁻².

29. SARA-ICE PoDs, computed as the mean of the ED₀₁ distribution conditional on it being less than 60,000 µg cm⁻², were compared to reference DSA1+ values⁸ reported in literature (18). The reference value is termed the *median sensitization potency estimate* (MSPE) and is reported as a weight-of-evidence potency metric calculated from human data. SARA-ICE PoD estimates were computed for the 196 chemicals in the defined approaches for skin sensitisation (DASS) reference data (12) using a) NAM data also reported in the evaluation set (single input for DPRA, KeratinoSens and h-CLAT) and b) a single kDPRA sourced from (Natsch and Gerberick, 2022). A DSA1+ MSPE value was defined for 54 / 66 chemicals with human reference data in the DASS evaluation set. A positive correlation exists (Pearson correlation coefficient is 0.64) and 46 / 54 (85%) of SARA-ICE PoDs are within 10-fold of the reference value. Roughly one third, 19 / 54 (35%) of SARA-ICE PoDs are within 2-fold of the reference value. SARA-ICE PoDs, calculated in the same way as above, were also compared to 154 chemicals with reference LLNA MLLPs sourced from OECD DASS reference data (12). A positive correlation exists (Pearson correlation coefficient is 0.71) and 122 / 154 (79%) of SARA-ICE PoDs are within 10-fold of the reference value. Again, roughly one third, 49 / 154 (32%) of SARA-ICE PoDs are within 2-fold of the reference value. Figure 1.1 shows SARA-ICE PoDs compared to reference MSPEs (left) and reference LLNA MLLPs (right). *Further analyses are made in Appendix 3 of the Supporting document to the Guideline (GL) on Defined Approaches (DAs) for Skin Sensitisation (12).*

⁷ <https://ntp.niehs.nih.gov/whatwestudy/niceatm/test-method-evaluations/skin-sens/da/SARA-ICE>

⁸ The hypothetical dose per skin area (DSA) that sensitizes exactly one test subject

Figure 1.1. Comparison of SARA-ICE PoDs estimated vs. reference values



Left: Comparison of SARA-ICE PoDs estimated from *in chemico* and *in vitro* inputs against reference DSA1+ MSPE values. Right: Comparison of SARA-ICE PoDs against reference LLNA MLLPs. Red points indicate compounds which are negative in the LLNA. The grey zone indicates the region in which PoDs and reference values are within 10-fold of each other.

1.3 Limitations

30. The identified limitations of the DAs and their individual components are summarised below. The limitations of the DAs are based on the limitations of the individual *in chemico/in vitro/in silico* information sources. Details on the limitations of individual *in chemico/in vitro* and *in silico* information sources to determine confidence in Part I/Part II DA predictions are provided in **Sections 2.1.4** and **3.1.4** and in the respective TGs (TG 442C, Appendices I-II; TG 442D, Appendices IA-IC; TG 442E, Annexes 1-4) (7, 8, 9) and in Appendices I-II of this Guideline.

31. The SARA-ICE DA is intended to predict a point of departure for use in risk assessment, so the performance assessment is based on quantitative regression (Figure 1.1). More details are provided in **Part III** of this Guideline, as well as in the *Supporting Document to the Guideline (GL) on Defined Approaches (DAs) for Skin Sensitisation (12)*.

1.3.1 Limitations of individual *in chemico/in vitro* information sources

32. Users should refer to the limitations of the individual *in chemico/in vitro* test methods as specified in their respective TGs, which are revised as new data become available and should be consulted regularly (TG 442C, Appendices I-II; TG 442D, Appendices IA-IC; TG 442E Annexes 1-4) (7, 8, 9). The most up-to-date published version of the respective TGs should always be used. For example, some types of chemicals such as metals, inorganic compounds, lipophilic substances, substances of unknown or variable compositions, complex reaction products and biological materials (UVCBs) and mixtures, may not be within the applicability domain for certain test methods. When considering testing of mixtures, difficult-to-test chemicals (e.g. unstable), or test chemicals not clearly within the applicability domain, upfront consideration should be given to whether the results of such testing will yield results that are meaningful scientifically. In cases where evidence can be demonstrated on the non-applicability of the information

source to specific categories of chemicals, the information source should not be used for those specific categories of chemicals.

33. Individual assay results within borderline ranges (BRs; **Annex 1**) may yield inconclusive 2o3 DA predictions. The applicability of individual information sources is described in the relevant Test Guideline (*in chemico/in vitro*). The consideration of limitations of individual *in chemico/in vitro* test methods in each 2o3 and ITS DA is detailed in **Section 2.1.4 (Figure 2.1)** and **Section 3.1.4 (Figure 3.1)**.

1.3.2 Limitations of *in silico* information sources

34. The ITS DA includes an *in silico* tool as an information source. These tools can either perform automated read-across or (Q)SAR predictions. (Q)SARs include both structure- activity relationship (SAR) models (*i.e.* structural alerts, expert systems) and quantitative structure-activity relationship (QSAR) models (*i.e.* statistical tools). (Q)SAR models should fulfil the OECD Principles for the Validation, for Regulatory Purposes, of (Q)SAR Models and be described in a QSAR Model Reporting Format (QMRF) document (16, 17)⁹. One of the OECD QSAR validation principles refers to a defined domain of applicability. The applicability domain is the response and chemical structure space in which the model makes predictions with a given reliability and the applicability domain should therefore consider the parametric, structural, mechanistic, metabolic and response space of the model. Where a DA for skin sensitisation includes an *in silico* tool, users should refer to the limitations and applicability domain of the individual *in silico* tool as detailed in **Appendices I-II** of this Guideline. Depending on the DIP, chemicals outside the *in silico* model's applicability domain may result in DA predictions of low confidence that are considered inconclusive.

1.3.3 Limitations of DAs

35. For the 2o3 DA, BRs have been defined for the individual assays addressing the three KE of the DA, in order to define areas where lower confidence may exist (for details see **Section 2.1.4** and **Annex 1** of this Guideline, and **Section 3.3** and **Annex 7** of the *Supporting document to the Guideline (GL) on Defined Approaches (DAs) for Skin Sensitisation*) (12). Positive and/or negative test results falling within these BRs and/or individual assay limitations have lower confidence and may result in inconclusive 2o3 DA predictions. Inconclusive predictions are also possible for the ITS DA (for details see **Section 3.1.4 (Figure 3.1)** and **Annex 2** of this Guideline. Inconclusive predictions are not relevant for SARA-ICE for PoD determination as this is a continuous metric (not categorical) made upon the assumption a chemical is a sensitiser.

36. Inconclusive DA predictions may nevertheless be considered in a weight-of- evidence approach and/or within the context of an IATA together with other information sources (*e.g.* demonstration of exposure to the test system, existing *in vivo* data, clinical data, read-across, other *in vitro/in chemico/in silico* data, etc.).

37. As the available data for each method and *in silico* tool differ slightly, maximal common datasets were used for each permutation to allow direct comparison to the performance of the classic DAs, *i.e.* 2o3 (original: DPRA, KeratinoSens™, h-CLAT) and ITS (original: DPRA, h-CLAT, Derek Nexus (ITSv1)/OECD (Q)SAR Toolbox (ITSv2)). The maximal common datasets use all available data points for a specific permutation of the DA. Beyond specific method or tool applicability domain limitations, no notable differences in performance among the DAs, including substitution of original with alternate information

⁹ The QMRF has been slightly adapted for reporting other *in silico* model predictions in the context of DASS. The adapted QPRF can be found on the OECD site for spreadsheets and software associated with OECD Test Guidelines on Health Effects: <https://www.oecd.org/en/topics/sub-issues/testing-of-chemicals/software-test-guidelines.html>

sources based on similar methods¹⁰, were noted based on specific physicochemical properties or chemical classes. Hazard identification performance of the DAs in comparison to LLNA-based reference data is broadly equivalent to that of the classic DAs. Further, for hazard, all DAs offer better performance than the LLNA against human-based reference data. For potency, when compared against human-based reference data, 8/10 (80%) ITS permutations have equivalent or better accuracy than the LLNA. (60%). Two ITS permutations (**Table 1.3**) had lower accuracy than the LLNA when compared to human-based reference data which was due to several underpredictions of 1A sensitisers as 1B sensitisers. These underpredictions may be attributed to several considerations, including human-based sub-categorizations derived from human data where the predicted sensitising dose was close to the threshold between GHS 1A and GHS 1B sub-categorizations (termed GHS 1A- in **Annex 4** of the *Supporting document to the OECD Guideline 497 on Defined Approaches for Skin Sensitisation* for more information on how this was derived). This lower accuracy is specific to these ITS DA permutations only and should be taken into consideration when selecting specific DAs to use; use of these specific iterations needs to be justified. These underprediction cases are discussed further in Annex 9 of the *Supporting document to the Guideline (GL) on Defined Approaches (DAs) for Skin Sensitisation* (1). In addition, in determining which DA to use, the applicability domain of the substance should be the primary driver on prioritising the selection of a method from TG 442C-E and/or *in silico* supporting documentation (Appendices I-II of this Guideline). DAs that utilise methods outside the applicability domain of the test substance are not appropriate for use. Additional considerations in selecting a DA may be related to the ultimate regulatory need.

1.4 References

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¹⁰ Similar methods in this guideline refer to methods that target a same key event (KE) and/or that are based on a similar technology.

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10. OECD (2017), *Guidance Document on the Reporting of Defined Approaches and Individual Information Sources to be Used within Integrated Approaches to Testing and Assessment (IATA) for Skin Sensitisation*, OECD Series on Testing and Assessment, No. 256, OECD Publishing, Paris, <https://doi.org/10.1787/9789264279285-en>.
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Part I SECTION 2 - Defined Approaches for Skin Sensitisation Hazard Identification

38. Part I of this guideline applies to Defined Approaches (DAs) that are intended solely for hazard identification, *i.e.* distinguishing between sensitisers and non-sensitisers. A summary of the DAs for hazard identification is provided below; additional detailed information can be found in the *Supporting document to the Guideline (GL) on Defined Approaches (DAs) for Skin Sensitisation* (1). Note that the ITS DAs can also be used for hazard characterization; please see **Part II** for more detail.

2.1 “2 out of 3” Defined Approach

2.1.1 Summary

39. The 2 out of 3 (2o3) DA is intended for the identification of the skin sensitisation hazard of a chemical (*i.e.* UN GHS Cat. 1 vs. UN GHS NC) without the use of animal testing. The data interpretation procedure (DIP) is currently not designed to provide information on the potency of a skin sensitiser.

40. The combination of test methods included in the 2o3 DA covers at least two of the first three KEs of the AOP leading to skin sensitisation, *i.e.* KE1: protein binding (OECD TG 442C) (2); KE2: keratinocyte activation (OECD TG 442D) (3); and KE3: dendritic cell activation (OECD TG 442E) (4).

41. The DIP requires that two concordant results obtained from methods addressing at least two of the first three KEs of the skin sensitisation AOP determine the final classification. For the classic 2o3 DA was evaluated against a curated dataset of up to 168 chemicals with LLNA-based reference data and up to 66 chemicals with human-based reference data agreed upon by the EG DASS (**Annex 2** of the *Supporting Document to the Guideline (GL) on Defined Approaches (DAs) for Skin Sensitisation* (1)). The performance of the DAs using the original (*i.e.*, classic DAs) and alternative information sources has been evaluated against both LLNA and human-based reference data, according to available datasets for each permutation, and can be found in **Tables 2.2-2.4** below and in **Annexes 2 and 9** of the *Supporting Document to the Guideline (GL) on Defined Approaches (DAs) for Skin Sensitisation* (1). It should be noted that due to the imbalanced nature of the reference data (higher numbers of UN GHS Cat. 1 than UN GHS NC), the measures of balanced accuracy are more uncertain, particularly in the case of the human data comparison.

2.1.2 Data interpretation procedure

42. The DIP in the 2o3 DA is a transparent, rule-based approach requiring no expert judgment (5, 6, 7). The approach predicts skin sensitisation hazard by sequential testing, in an undefined order, in up to three of the internationally accepted non-animal assays mapping to KE1-3. In other words, the assays are run for two KEs, and if these assays provide consistent results, then the chemical is predicted accordingly as skin sensitiser or non-sensitiser. If the first two assays provide discordant results, the assay for the remaining KE is run. The overall result is based on the two concordant findings considering the confidence of the obtained predictions as described in **Section 2.1.4**.

43. The performance of the 2o3 DA was found to be impacted by the consideration of BRs for each of the methods, as described below in **Section 2.1.4**, and further detailed in **Section 3.3** and **Annex 7** of the *Supporting Document to the Guideline (GL) on Defined Approaches (DAs) for Skin Sensitisation* (1). A decision tree is provided in **Figure 2.1** of **Section 2.1.4** to derive predictions for the 2o3 DA, with no modification of the 2o3 DA DIP.

2.1.3 Description and limitations of the individual information sources

44. The individual information sources in the DA are assays included in OECD KE- based TGs for skin sensitisation (OECD TG 442C, 442D, 442E) (2, 3, 4), and the details for the individual protocols are provided therein.

45. The following assays from those TGs have been characterised and included in the 2o3 DA.

- KE1 (Amino acid Derivative Reactivity Assay (ADRA), Direct Peptide Reactivity Assay (DPRA); TG 442C) (2): Skin sensitisers are generally electrophilic and react with the nucleophilic moieties of proteins. The ADRA and DPRA measure depletion of two peptides/amino acids containing either cysteine or lysine residues due to covalent binding. Test protocols and prediction models are found in the TG. If borderline results are obtained for peptide depletion, additional testing should be conducted, as specified in OECD TG 442C and in **Annex 1**.
- KE2 (EpiSensA, LuSens, KeratinoSens™; TG 442D) (3): Activation of keratinocytes is demonstrated via either a) reaction of human, immortalized keratinocytes harbouring a reporter gene construct via the Nrf2-Keap1 pathway (KeratinoSens, LuSens), or b) quantification of changes in the expression of marker genes associated with keratinocyte activation in a reconstructed human epidermis model (EpiSensA). Test protocols and prediction models are found in the TG. If borderline results are obtained for luciferase induction, additional testing should be conducted, as specified in **Annex 1**.
- KE3 (Genomic Allergen Rapid Detection test (GARD@Skin), Human Cell Line Activation Test (h-CLAT), Interleukin-8 Reporter Gene Assay (IL-8 Luc), U937 Cell Line Activation Test (U-SENS); TG 442E) (4): Activation of antigen presenting cells is characterised by either a) a change in the expression of cell surface marker(s) associated with the process of activation of monocytes and DC following exposure to sensitisers (e.g. CD86, CD54; h-CLAT, U-SENS), b) changes in IL-8 expression, a cytokine associated with the activation of DC (IL-8 Luc) or c) an end-point specific genomic biomarker signature (GARD@skin). Test protocols and prediction models are found in the TG. If borderline results are obtained for the assays, additional testing should be conducted, as specified in **Annex 1**.

46. The current limitations of individual *in chemico* and *in vitro* test methods, such as limited solubility, are described in their respective TGs (TG 442C, Appendices I-II; TG 442D, Appendices IA-I1C; TG 442E, Annexes 1-4) and the validation studies cited therein (2, 3, 4).

2.1.4 Confidence in the 2o3 DA predictions

47. The limitations of the *in chemico* and *in vitro* methods used in the 2o3 DA dictate whether test results can be used to support hazard identification and are reported in their respective TGs [TG 442C, TG 442D, TG 442E] (e.g. for substances that do not provide conclusive results in the individual methods due to solubility reasons) (2, 3, 4). Additionally, assay results are subject to variability (due to, for example, minor fluctuations in cell response) and these variations increase the uncertainty of a test result especially when close to a (classification) cut-off. In order to define areas where lower confidence in the DA results may exist, borderline ranges (BRs) have been statistically derived for each assay addressing the three KE of the 2o3 DA, (see **Annex 1** of this document, and **Section 3.3** and **Annex 7** of the *Supporting document to the Guideline (GL) on Defined Approaches (DAs) for Skin Sensitisation*) (1). The specific BRs for each assay, as derived from their respective validation study data, are summarised in **Table 2.1**.

Table 2.1. Borderline ranges for each assay defined for use in the 2o3 DA

Method	Parameter: Borderline Range
ADRA	4.1% ≤ Mean percent depletion: - ≤ 5.9% or 4.7% ≤ NAC-only depletion ≤ 6.8%
DPRA	4.95% ≤ Mean peptide depletion ≤ 8.32% or 10.56% ≤ Cys-only depletion ≤ 18.47%
KeratinoSens™	1.35-fold ≤ Luciferase induction ≤ 1.67-fold
LuSens EpiSensA	1.28-fold ≤ Luciferase induction ≤ 1.76 10.71-fold ≤ I _{max} for ATF3 ≤ 21.02 1.64-fold ≤ I _{max} for GCLM ≤ 2.45 1.61-fold ≤ I _{max} for DNAJB2 ≤ 2.52 3.11-fold ≤ I _{max} for IL-8 ≤ 5.16
GARD@skin	Mean DV: -0.450 - +0.450
h-CLAT	157% ≤ RFI CD54 ≤ 255% 122% ≤ RFI CD86 ≤ 184%
IL-8 Luc	1.25 ≤ Ind-IL8LA ≤ 1.57
U-SENS™	128% ≤ SI CD86 ≤ 176%

Note: BR values were derived according to the methodologies described in (8-11). See Annex 1 for decision tree and Annex 7 (Supplemental Materials) for additional derivation information.

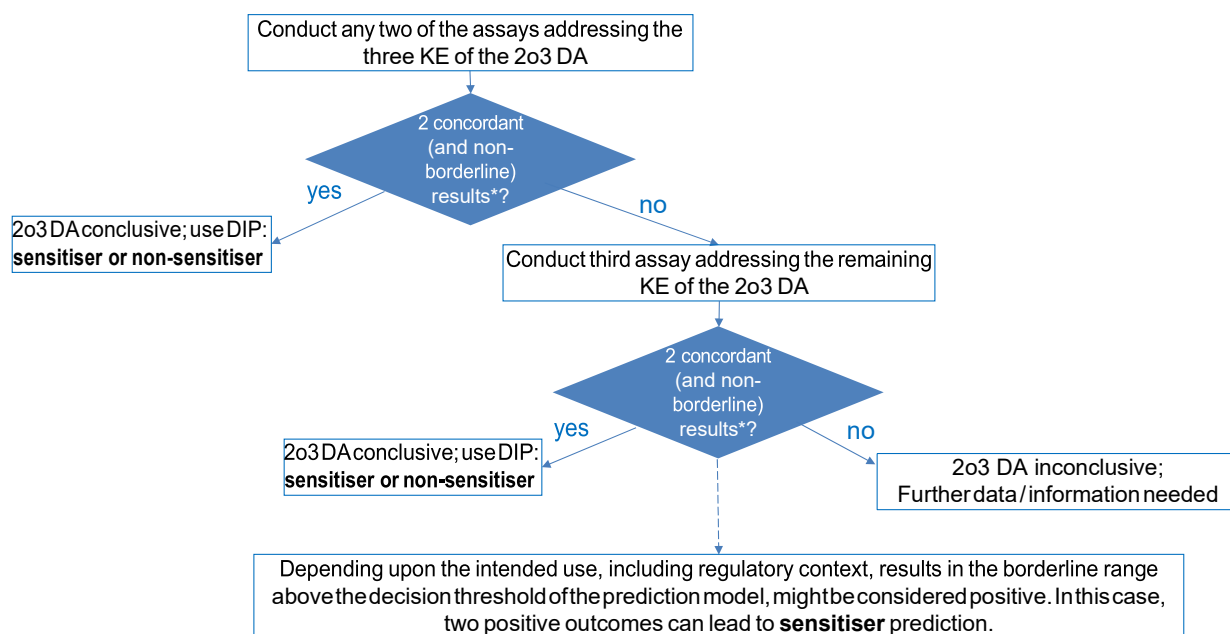
48. The incorporation of BRs into the prediction models (PM) for each of the individual information sources is described in detail in **Annex 1** of this guideline.

49. Positive and negative test results falling within these BRs as well as inconclusive results due to limitations in the *in chemico/in vitro* TGs are of lower confidence, and affect the outcome of the 2o3 DA as described below:

- In case the result of one of the 2o3 DA test methods falls into the respective test method's BR, a 2o3 DA prediction can still be made if the outcomes of the other two test methods composing the 2o3 DA are concordant and have high confidence (*i.e.*, results falling outside of the respective BRs).
- Similarly, in case a result is obtained in one test method that is considered inconclusive due to limitations or the chemical falling outside the applicability domain of the method, a 2o3 DA prediction can still be made if the outcomes of the other two test methods composing the 2o3 DA are concordant and have high confidence (*i.e.*, results falling outside of the respective BRs).
- However, if the result of one of the 2o3 DA test methods falls into the respective test method's BR or limitation, and the other two methods composing the 2o3 do not provide concordant and high confidence results, the 2o3 DA prediction is considered 'inconclusive'. These inconclusive predictions may nevertheless be considered in a weight-of-evidence approach and/or within the context of an IATA together with other information sources. Depending on the intended use, including regulatory context, results in the borderline range above the decision threshold of the prediction model might still be considered positive; in this case, two positive outcomes can lead to an overall positive (sensitiser) prediction.

50. These borderline considerations and their impact on the confidence of the 2o3 DA predictions are visualised in **Figure 2.1**. DA predictions with high confidence for hazard identification are considered conclusive. DA predictions with low confidence are considered inconclusive for hazard identification. These 'inconclusive' predictions may nevertheless be considered in a weight-of-evidence approach and/or within the context of an IATA together with other information sources.

Figure 2.1 Decision tree to be used for the 2o3 DA, taking into account borderline result



Note: *Borderline results are determined based on workflows given in Annex 1 and assay limitations are described in their respective test guidelines (TG 442C, Appendices I-II; TG 442D, Appendices IA-IC; TG 442E, Annexes 1-4). DA = defined approach; DIP = data interpretation procedure; KE = key event; 2o3 = 2 out of 3 DA.

2.1.5 Predictive capacity of the 2o3 DA vs. the LLNA-based reference data

51. The predictive capacity of the 2o3 DA is reported based on data generated by the LLNA (see **Table 2.2** and **Annex 3** of the *Supporting document to the Guideline (GL) on Defined Approaches (DAs) for Skin Sensitisation*) (1). The BR analyses were applied as described above to assign confidence to the 2o3 DA predictions. Performance statistics are reported for conclusive (high confidence) predictions as compared to LLNA-based reference data. Detailed performance statistics and further details for each permutation of the 2o3 DA are available in **Annexes 2** and **9** of the *Supporting Document to the Guideline (GL) on Defined Approaches (DAs) for Skin Sensitisation* (1).

Table 2.2. Hazard identification performance of the 2o3 DA in comparison to LLNA reference data

DA Performance (N=74-129)	
Accuracy (%)	81-95%
Sensitivity (%)	82-97%
Specificity (%)	72-100%
Balanced Accuracy (%)	80-95%

Note: (i) Accuracy is the correct classification rate, sensitivity is the true positive rate, specificity is the true negative rate, and balanced accuracy is the average of sensitivity and specificity with respect to LLNA reference data. Statistics reflect conclusive predictions only. Additional performance characterisation is available in the *Supporting document to the Guideline (GL) on Defined Approaches (DAs) for Skin Sensitisation*. (ii) the full range of performance is shown here for transparency, and the broad range is attributed to different data availability for each permutation of 2o3 DA. However, it should be noted that the equivalent performance of all methods when substituted in 2o3 DA for the same set of chemicals as the classic 2o3 DA has been demonstrated (for more information see **Annex 2** and **Annex 9**)

52. The application of the BR analyses and the designation of high/low confidence for the 2o3 DA predictions are applied as described above in **Section 2.1.4** and **Annex 1**, and further detailed in **Section 3.3** and **Annex 7** of the *Supporting document to the Guideline (GL) on Defined Approaches (DAs) for Skin Sensitisation* (1).

53. Due to the imbalanced nature of the reference data, the measure of specificity (based on 9-25 LLNA negative chemicals) is more uncertain than the measure of sensitivity (based on 65-105 LLNA positive chemicals).¹¹

2.1.6 Predictive capacity of the 2o3 DA vs. Human Data

54. The predictive capacity of the 2o3 DA is also reported based on human data (see **Table 2.3**). The BR analyses were applied as described above to assign confidence to the 2o3 DA predictions. Performance statistics are reported for conclusive (high confidence) predictions as compared to human-based reference data. Detailed performance statistics for each permutation of the 2o3 DA are available in predictions for specific chemicals and further details are available in **Section 5** and **Annexes 2 and 9** of the *Supporting document to the Guideline (GL) on Defined Approaches (DAs) for Skin Sensitisation* (1).

¹¹ Because the balance of positive and negative LLNA chemicals differs depending on the DA permutation assessed, the sum of the range of positive and negative chemicals will not equal the total "N" in Table 2.2.

Table 2.3. Hazard identification performance of the 2o3 DA in comparison to human-based reference data

DA Performance vs. Human Data (N=35-55)	
Accuracy (%)	83-95%
Sensitivity (%)	84-97%
Specificity (%)	50-100%
Balanced Accuracy (%)	72-97%

Note: (i) Accuracy is the correct classification rate, sensitivity is the true positive rate, specificity is the true negative rate, and balanced accuracy is the average of sensitivity and specificity with respect to human data. Statistics reflect conclusive predictions only. Additional performance characterisation is available in the *Supporting document to the Guideline (GL) on Defined Approaches (DAs) for Skin Sensitisation* (1). (ii) the full range of performance is shown here for transparency, and the broad range is attributed to different data availability for each permutation of 2o3 DA. However, it should be noted that the equivalent performance of all methods when substituted in 2o3 DA for the same set of chemicals as the classic 2o3 DA has been demonstrated (for more information see **Annex 2** and **Annex 9**)

55. The application of the BR analyses and the designation of high/low confidence for the 2o3 DA predictions are applied as described above in **Section 2.1.4** and **Annex 1**, and further detailed in **Section 3.3 and Annex 7** of the *Supporting document to the Guideline (GL) on Defined Approaches (DAs) for Skin Sensitisation* (1).

56. Due to the imbalanced nature of the reference data, the measure of specificity (based on 4-8 human negative chemicals) is more uncertain than the measure of sensitivity (based on 30-45 human positive chemicals).¹²

2.1.7 Predictive capacity of the LLNA vs. Human Data

57. To provide a basis for comparison for the DA performance statistics given above, the predictive capacity of the LLNA is reported based on human data (see **Table 2.4**). Data for specific chemicals and further details are available in **Section 5** and **Annex 2** of the *Supporting document to the Guideline (GL) on Defined Approaches (DAs) for Skin Sensitisation* (1).

Table 2.4. Hazard identification performance of the LLNA in comparison to Human-based reference data

LLNA Performance vs. Human Data (N=56)	
Accuracy (%)	82%
Sensitivity (%)	94%
Specificity (%)	22%
Balanced Accuracy (%)	58%

Note: Accuracy is the correct classification rate, sensitivity is the true positive rate, specificity is the true negative rate, and balanced accuracy is the average of sensitivity and specificity with respect to human data. Additional performance characterisation is available in the *Supporting document to the Guideline (GL) on Defined Approaches (DAs) for Skin Sensitisation* (1).

¹² Because the balance of positive and negative LLNA chemicals differs depending on the DA permutation assessed, the sum of the range of positive and negative chemicals will not equal the total "N" in Table 2.3.

58. The hazard identification performance of the conclusive 2o3 DA predictions vs. human data was 83-95% accuracy, 84-97% sensitivity, 50-100% specificity, and 72-97% balanced accuracy, comparable to or exceeding the performance of the LLNA- vs human-based data.

59. As previously noted, due to the imbalanced nature of the reference data, the measures of specificity are more uncertain than the measures of sensitivity.

2.1.8 Proficiency chemicals

60. The 2o3 DA relies on a simple, rule-based data interpretation procedure and requires no expert judgment. Proficiency chemicals for the individual information sources (KE1-3) are defined in the respective guidelines (2, 3, 4). Proficiency for the individual information sources demonstrates proficiency for the DA.

2.1.9 Reporting of the DA

61. The reporting of the DA application should follow the template described in OECD GD 255 (12), and should include at a minimum the following elements:

- Test chemical identification (e.g. chemical name, structural formula, composition, isomers, impurities including their quantities as available, CAS number, batch and lot number, and other relevant identifiers)
- Indicate the DA and justify the specific information sources used
- Individual test reports performed per corresponding guideline (OECD TG 442C, 442D, 442E). Note that the chemical identity for each test report should match that above
- Application of the individual prediction models adapted to be used within the 2o3 DA to determine borderline outcomes, as described in **Annex 1**
- Outcome of the DA application (hazard identification, i.e. skin sensitiser or not skin sensitiser or inconclusive result)
- Any deviation from or adaptation of the 2o3 DA
- Conclusion

2.2 References

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62. **Part II** of this Guideline includes Defined Approaches that allow the allocation of skin sensitisers into UN GHS sub-category 1A, strong sensitisers, or sub-category 1B for other (moderate to weak) skin

Part II SECTION 3 - Defined Approaches for Skin Sensitisation Potency Categorisation

sensitisers. These DAs may also be used to distinguish between sensitisers (UN GHS Category 1) and non-sensitisers (no classification; NC) for hazard identification. Additional detailed information can be found in the *Supporting document to the Guideline (GL) on Defined Approaches (DAs) for Skin Sensitisation* (1).

3.1 “Integrated Testing Strategy (ITS)” Defined Approach

3.1.1 Summary

63. This defined approach was constructed as an Integrated Testing Strategy (ITS) for prediction of the skin sensitisation hazard potential and potency sub-categorisation according to the UN GHS (sub-categories 1A and 1B) classification system.

64. The ITS DA uses test methods that address key events (KEs) 1 and 3 in the Adverse Outcome Pathway (AOP) leading to skin sensitisation as: KE1: protein binding (TG 442C) (2); KE3: dendritic cell activation formally described by the OECD (TG 442E) (3) and includes an *in silico* prediction of skin sensitisation. The *in silico* prediction of skin sensitisation is provided by Derek Nexus (4) or the OECD QSAR Toolbox (5).

65. The DIP requires a score to be calculated from each information source based on the assay-specific thresholds and the total score used to determine the final classification (**Table 3.1**). the classic ITS DA was evaluated for hazard and potency identification against a curated dataset of up to 168 chemicals with LLNA-based reference data and up to 66 chemicals with human-based reference data as found in **Annex 2** of the *Supporting document to the Guideline (GL) on Defined Approaches (DAs) for Skin Sensitisation* (1). The performance of the DAs containing alternate information sources was evaluated against between 72-159 and 37-64 substances with both LLNA-based and human-based reference data, respectively. When compared against the LLNA- and human-based reference data for hazard, the ITS DA demonstrated balanced accuracies in the range of 78-88% and 67-76%, comparable to the classic ITS DA (see **Table 3.2** and **Table 3.4**). The ITS DA demonstrated accuracies in the range of 66-80% when evaluated against LLNA-based potency data (see **Table 3.3**). When evaluated against the human data for potency, the ITS DA demonstrated comparable accuracy to the LLNA, with a range of 54-75% (vs 60% for the LLNA) (see **Table 3.5** and **3.7**). When compared against human-based reference data, 8/10 (80%) ITS permutations have equivalent or better accuracy than the LLNA. (60%). Two ITS permutations (**Table 1.3**) had lower accuracy than the LLNA when compared to human-based reference data which was due to several underpredictions of 1A sensitisers as 1B sensitisers. These underpredictions may be attributed to several considerations, including human-based sub-categorizations derived from human data where the predicted sensitising dose was close to the threshold between GHS 1A and GHS 1B sub-categorizations (termed GHS 1A-) in **Annex 4** of the *Supporting document to the OECD Guideline 497 on Defined Approaches for Skin Sensitisation* for more information on how this was derived). This lower accuracy is specific to these ITS DA permutations only and should be taken into consideration when selecting specific DAs to use; selection of these specific DAs needs to be justified. These underprediction cases are discussed further in Annex 9 of the *Supporting document to the Guideline (GL) on Defined Approaches (DAs) for Skin Sensitisation* (1).

3.1.2 Data interpretation procedure

66. The ITS DIP uses scores assigned to the quantitative results from a KE3 (3) and a KE1 assay (1), and from an *in silico* tool to discriminate chemicals into UN GHS category 1A (strong sensitiser); category 1B (other sensitiser), or Not Classified (non-sensitiser) (**Table 3.1**).

67. The DIP was amended from the original published version of the ITS (6-7) to change the cut-off for 1A sensitisers from a score of 7 to a score of 6 to optimize the ability of the DA to detect strong sensitisers and to extend the applicability of the ITS to chemicals for which *in silico* predictions cannot be generated. The DIP was also altered from the published version in that it was originally applied to ECETOC categories¹³, and is here applied to the UN GHS subcategories.

68. The quantitative results of the KE1 and KE3 assays are converted into a score from 0 to 3 for each assay, as shown in **Table 3.1**. For the *in silico* prediction, a positive outcome is assigned a score of 1; a negative outcome is assigned a score of 0. Further details on the respective protocols are available in **Annexes 1 and 2** and **Appendices I-II**). When these scores have been assessed, a total battery score ranging from 0 to 7, calculated by summing the individual scores, is used to predict the sensitising potential (hazard identification; UN GHS Cat. 1 vs. UN GHS NC) and potency (UN GHS Cat. 1A, Cat. 1B and NC). The positive criteria for identifying skin sensitisers (UN GHS Cat. 1) are set as a total battery score of 2 or greater. Based on the updated DIP, a total battery score is assigned into three ranks: score of 6-7 is defined as a strong (UN GHS Cat. 1A) sensitiser; score of 2-5 as moderate/weak (UN GHS Cat. 1B) sensitiser; score of 1 or 0, as not classified (*i.e.* a non-sensitiser).

Table 3.1. Schematic of the ITS defined approach. The DA is a score-based system depending on KE1 and KE3 assays from TG 442E and 442C, and an *in silico* prediction, as shown.

Information Source Type	KE1 (OECD TG 442C)				KE3 (OECD TG 442E)			<i>In silico</i>
Information Sources Permitted	ADRA, DPRA				GARD@skin, h-CLAT, U-SENS TM			Derek Nexus, OECD QSAR Toolbox
Score	ADRA mean NAC & NAL % depletion	ADRA NAC % depletion*	DPRA mean Cys & Lys % depletion	DPRA Cys % depletion†	GARD @Skin (µg/mL)	h-CLAT MIT (µg/mL)	U-SENS TM EC150 (µg/mL)	Please refer to individual protocol to apply score
3	≥46.4	≥67.4	≥42.47	≥98.24	≤13.03	≤10	≤3	
2	≥15.5, <46.4	≥17.5, <67.4	≥22.62, <42.47	≥23.09, <98.24	>13.03, ≤56.44	>10, ≤150	>3, ≤35	
1	≥4.9, <15.5	≥5.6, <17.5	≥6.38, <22.62	≥13.89, <23.09	>56.44	>150, ≤5000	>35, ≤200	Positive
0	<4.9	<5.6	<6.38	<13.89	-	not calculated	not calculated / >200	Negative
Potency								
Total Battery Score								
UN GHS 1A 6-7								
UN GHS 1B 2-5								
Not classified 0-1								

Source: Adapted from Takenouchi (6) and Imamura (8)

Abbreviations: ADRA: Amino acid Derivative Reactivity Assay; NAC: N-(2-(1-naphthyl) acetyl)-L-cysteine, NAL: α-N-(2-(1-naphthyl) acetyl)-L-lysine; DPRA: Direct Peptide Reactivity Assay; Cys: cysteine, Lys: Lysine; GARD: Genomic Allergen Rapid Detection; h-CLAT: Human Cell Line Activation Test; MIT: minimum induction threshold; U-SENS: U-937 cell line activation test; EC: effective concentration.

¹³ ECETOC Technical Report 087 (2003), Contact Sensitisation: Classification According to Potency. Available at: <https://www.ecetoc.org/publication/tr-087-contact-sensitisation-classification-according-to-potency>

Note: UN GHS 1A correspond to strong sensitisers and UN GHS 1B correspond to other (moderate to weak) sensitisers. Not classified are considered non-sensitisers. For KE1, KE3, and *in silico*, only one assay is selected from each is used for determining a score. *NAC-only depletion thresholds are used in the case of co-elution with the NAL peptide. ‡Cysteine-only depletion thresholds are used in the case of co-elution with the lysine peptide.

3.1.3 Description and limitations of the individual information sources

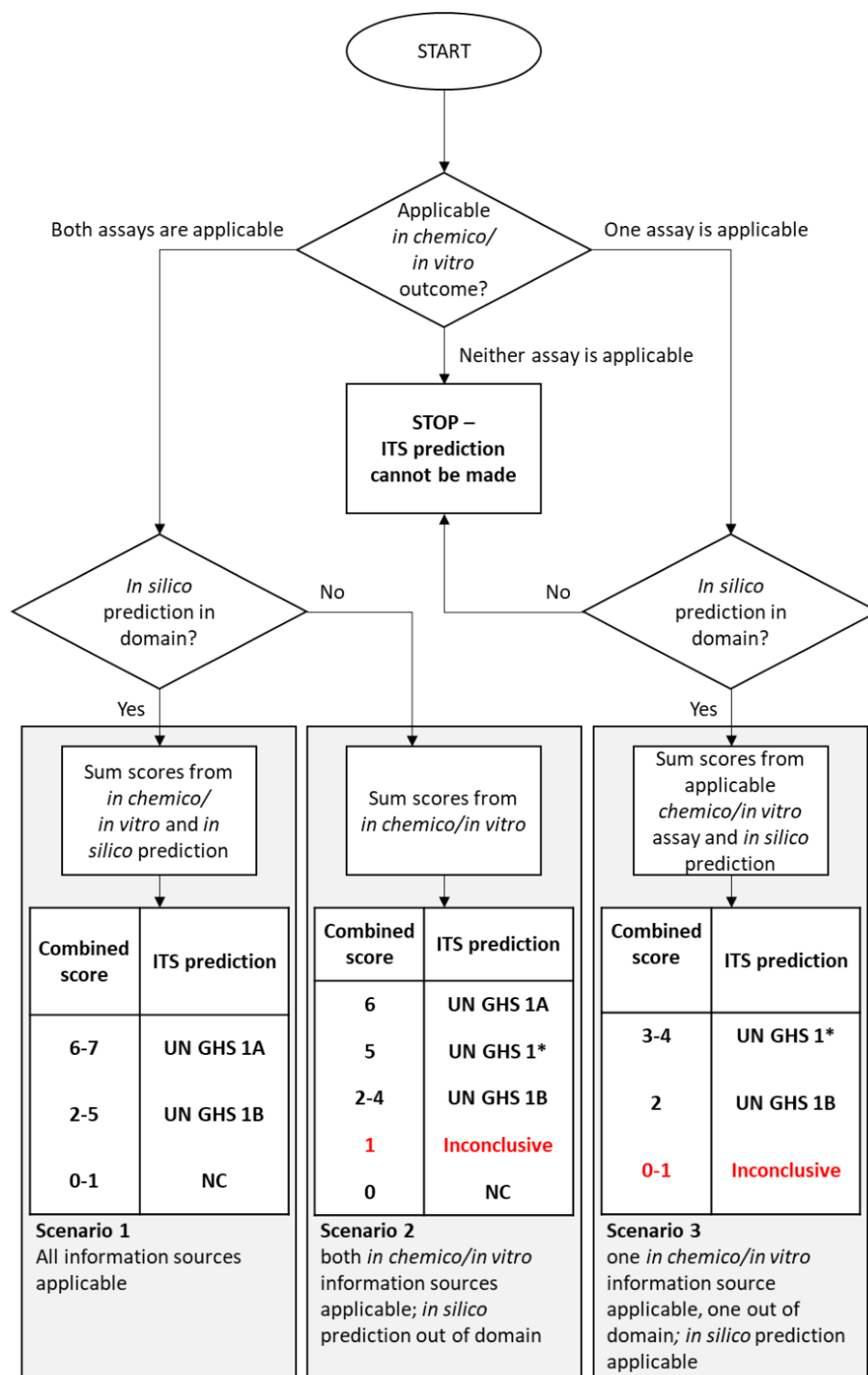
69. The individual *in chemico* and *in vitro* information sources are existing KE-based OECD TGs (TG 442C, 442E) (2, 3), and the protocols and limitations of each assay are detailed therein. For information on the *in silico* tools, including the description, limitations and applicability domain, users should refer to the individual relevant **Appendix**, as detailed in **Appendices I-II**.

3.1.4 Confidence in the ITS DA predictions

70. The level of confidence of the ITS DA prediction is assigned based on the total DA score and applicability domain of the individual information sources, as shown via the flow chart in Figure 3.1. The first decision on whether all information elements can be used is dictated by the limitations of the *in chemico* and *in vitro* methods, as found in TG 442C (2) and TG 442E (3), and by the applicability domain of the *in silico* prediction (**Annex 2**). Partial information sources (i.e. two *in chemico/in vitro* outcomes only, or one *in chemico/in vitro* outcome and an *in silico* prediction) may be used to obtain a DA prediction as shown via the flow chart in **Figure 3.1**.

71. DA predictions with high confidence for hazard identification and potency are considered conclusive. DA predictions with low confidence are considered inconclusive for hazard identification and/or potency. These ‘inconclusive’ predictions may nevertheless be considered in a weight-of-evidence approach and/or within the context of an IATA together with other information sources. Details including applicability domain and confidence considerations are provided in **Annex 2**.

Figure 3.1. Decision tree for assigning confidence to the ITS DA predictions



*Conclusive for hazard, inconclusive for potency

3.1.5 Predictive capacity of the ITS DA vs the LLNA-based reference data

72. The predictive capacity of the ITS DA is reported based on data from the LLNA (see **Tables 3.2-3.3**) (see **Section 1.1** and **Annexes 3** and **9** of the *Supporting document to the Guideline (GL) on Defined Approaches (DAs) for Skin Sensitisation*) (1). The workflow shown in **Figure 3.1** was applied to assign confidence to the ITS DA predictions. The designation of conclusive/inconclusive for the ITS DA predictions is further detailed in **Annex 2**. Performance statistics are reported for conclusive predictions as compared to LLNA-based reference data. DA predictions for specific chemicals and further details are available in **Section 5** and **Annexes 2** and **9** of the *Supporting document to the Guideline (GL) on Defined Approaches (DAs) for Skin Sensitisation* (1).

Table 3.2. Hazard identification performance of the ITS DA in comparison to LLNA reference data

ITS Performance vs. LLNA (N=122-159)	
Accuracy (%)	86-93%
Sensitivity (%)	91-94%
Specificity (%)	65-82%
Balanced Accuracy (%)	78-88%

Note: (i) Accuracy is the correct classification rate, sensitivity is the true positive rate, specificity is the true negative rate, and balanced accuracy is the average of sensitivity and specificity with respect to LLNA-based data. Statistics reflect high confidence predictions only. Additional performance characterisation is available in the *Supporting document to the Guideline (GL) on Defined Approaches (DAs) for Skin Sensitisation* (1). (ii) the full range of performance is shown here for transparency, and the broad range is attributed to different data availability for each permutation of ITS DA. However, it should be noted that the equivalent performance of all methods when substituted in ITS DA for the same set of chemicals as the classic ITS DA has been demonstrated (for more information see **Annex 2** and **Annex 9**).

73. Due to the imbalanced nature of the reference data, the measure of specificity (based on 11-30 LLNA negative chemicals) is more uncertain than the measure of sensitivity (based on 111-129 LLNA positive chemicals).¹⁴

Table 3.3. ITS DA vs. LLNA-based reference data: Statistics based on the UN GHS 1A/1B sub-categorisation

ITS Sub-categorisation Performance vs. LLNA (N=72-141)	
Correct classification (%)	66-80%
Underpredicted (%)	13-18%
Overpredicted (%)	4-17%

Note: Statistics reflect high confidence predictions only. Overprediction refers to chemicals misclassified as more potent; a NC predicted to be 1A/1B or a 1B sensitiser predicted to be a 1A. sensitiser. Underprediction refers to chemicals misclassified as less potent; a 1A sensitiser predicted to be a 1B/NC, a 1B sensitiser predicted to be NC. For more details on within-class performance, please see Annexes 2 and 9 of the *Supporting document to the Guideline (GL) on Defined Approaches (DAs) for Skin Sensitisation* (1).

¹⁴ Because the balance of positive and negative LLNA chemicals differs depending on the DA permutation assessed, the sum of the range of positive and negative chemicals will not equal the total "N" in **Tables 3.2** and **3.3**.

74. The designation of high/low confidence for the ITS DA predictions is applied as described above in **Figure 3.1** and further detailed in **Annex 2**.

3.1.6 Predictive capacity of the ITS DA vs Human Data

75. The predictive capacity of ITS DA is reported based on human data (see **Tables 3.4-3.5**), curated as agreed upon by the EG DASS. The designation of high/low confidence for the ITS DA predictions is further detailed in **Annex 2**. Performance statistics are reported for high confidence predictions as compared to human-based reference data, and inconclusive results are indicated. DA predictions for specific chemicals and further details are available in **Section 5** and **Annexes 2 and 9** of the *Supporting document to the Guideline (GL) on Defined Approaches (DAs) for Skin Sensitisation* (1).

Table 3.4. Hazard identification performance of the ITS DA in comparison to human-based reference data

ITS Performance vs. Human Data (N=52-64)	
Accuracy (%)	83-91%
Sensitivity (%)	87-96%
Specificity (%)	44-57%
Balanced Accuracy (%)	67-76%

Note: (i) Accuracy is the correct classification rate, sensitivity is the true positive rate, specificity is the true negative rate, and balanced accuracy is the average of sensitivity and specificity with respect to human data. Statistics reflect conclusive predictions only. Additional performance characterisation is available in the *Supporting document to the Guideline (GL) on Defined Approaches (DAs) for Skin Sensitisation* (1). (ii) The full range of performance is shown here for transparency, and the broad range is attributed to different data availability for each permutation of ITS DA. However, it should be noted that the equivalent performance of all methods when substituted in ITS DA for the same set of chemicals as the classic ITS DA has been demonstrated (for more information see **Annex 2** and **Annex 9**).

76. Due to the imbalanced nature of the reference data, the measure of specificity (based on 7-9 human negative chemicals) is more uncertain than the measure of sensitivity (based on 44-55 human positive chemicals).¹⁵

¹⁵ Because the balance of positive and negative LLNA chemicals differs depending on the DA permutation assessed, the sum of the range of positive and negative chemicals will not equal the total "N" in **Tables 3.4** or **3.5**.

Table 3.5. Potency categorisation performance of the ITS DA in comparison to human-based reference data, based on the UN GHS 1A/1B sub-categorisation

ITS Sub-categorisation Performance vs. Human Data (N=37-59)	
Correct classification (%)	54-75%
Underpredicted (%)	13-31%
Overpredicted (%)	13-18%

Note. Overprediction refers to chemicals misclassified as more potent; a NC predicted to be 1A/1B or a 1B sensitiser predicted to be a 1A. Underprediction refers to chemicals misclassified as less potent; a 1A sensitiser predicted to be a 1B/NC, a 1B sensitiser predicted to be NC. For more details on within- class performance, please see Annexes 2 and 9 of the *Supporting document to the Guideline (GL) on Defined Approaches (DAs) for Skin Sensitisation* (1).

77. The designation of conclusive/inconclusive for the ITS DA predictions is applied as described above in **Figure 3.1** and further detailed in **Annex 2**.

78. Due to the imbalanced nature of the reference data and the small numbers of chemicals, the measures of specificity, and consequently also balanced accuracy, are more uncertain for classes that contain a low number of chemicals (e.g. for NC chemicals).

79. The hazard identification performance of the conclusive ITS DA predictions vs. human data was 83-91% accuracy, 87-96% sensitivity, 44-57% specificity, and 67-76% balanced accuracy, comparable to or exceeding the performance of the LLNA (**Table 3.4**).

80. As previously noted, due to the imbalanced nature of the reference data, the measures of specificity are more uncertain than the measures of sensitivity.

81. The performance of the conclusive ITS DA predictions vs. human data for potency sub-categorisation showed 54-75% correct classification overall, with overpredictions between 13-18% and underpredictions of 13-12%, comparable to or exceeding the performance of the LLNA (**Table 3.5**).

82. As previously noted, due to the imbalanced nature of the reference data and the small numbers of chemicals, the measures of specificity, and consequently also balanced accuracy, are more uncertain for classes that contain a low number of chemicals, e.g. for NC chemicals.

3.1.7 Predictive capacity of the LLNA vs. Human Data

83. To provide a basis for comparison for the DA performance, the predictive capacity of the LLNA is reported based on data from the Human Predictive Patch Test (HPPT) (see **Table 3.6** and **3.7**). Data for specific chemicals and further details are available in **Section 5** and **Annex 2** of the *Supporting document to the Guideline (GL) on Defined Approaches (DAs) for Skin Sensitisation* (1).

Table 3.6. Hazard identification performance of the LLNA in comparison to human-based reference data

LLNA Hazard ID Performance vs. Human Data (N=56)	
Accuracy (%)	82%
Sensitivity (%)	94%
Specificity (%)	22%
Balanced Accuracy (%)	58%

Note: Accuracy is the correct classification rate, sensitivity is the true positive rate, specificity is the true negative rate, and balanced accuracy is the average of sensitivity and specificity with respect to HPPT-based data. Additional performance characterisation is available in the *Supporting document to the Guideline (GL) on Defined Approaches (DAs) for Skin Sensitisation (1)*.

Table 3.7. Potency categorisation performance of the LLNA in comparison to human-based reference data, based on the UN GHS 1A/1B sub-categorisation

LLNA Sub-categorisation Performance vs. Human Data (N=47)	
Correct classification (%)	60%
Underpredicted (%)	21%
Overpredicted (%)	19%

Note: Overprediction refers to chemicals misclassified as more potent; a NC predicted to be 1A/1B or a 1B sensitiser predicted to be a 1A. Underprediction refers to chemicals misclassified as less potent; a 1A sensitiser predicted to be a 1B/NC, a 1B sensitiser predicted to be NC. For more details on within- class performance, please see Annexes 2 and 9 of the *Supporting document to the Guideline (GL) on Defined Approaches (DAs) for Skin Sensitisation (1)*.

3.1.8 Proficiency chemicals

84. The ITS DA relies on a simple, rule-based data interpretation procedure and no expert judgment is required. Proficiency chemicals for the individual *in chemico* and *in vitro* information sources (KE1 and KE3) are defined in the respective guidelines (TG 442C, 442E) (2, 3). The protocol details for the *in silico* information source options are included in **Appendices I-II** of this guideline. Proficiency has been demonstrated for the *in silico* information source versions stated in the protocols in Appendices I-II by predicting the entire Annex 2 database (See Annex 9 of the *Supporting document to the Guideline (GL) on Defined Approaches (DAs) for Skin Sensitisation (1)*). More up to date versions may be used if proficiency matches or exceeds that of the original version. Proficiency for the individual information sources demonstrates proficiency for the DA.

3.1.9 Reporting of the DA

85. The reporting of the ITS DA should follow the template described in OECD GD 255 (9), and should include at a minimum the following elements:

- Test chemical identification (e.g. chemical name, structural formula, composition, isomers, impurities including their quantities as available, CAS number, batch and lot number, and other relevant identifiers).
- Indicate the DA and justify the specific information sources (methods and *in silico* model) used.

- Individual test reports for the individual tests performed per corresponding guideline (OECD TG 442C, 442E). Note that the chemical identity for each test report should match that above.
- Description of protocol used for *in silico* prediction (**Appendices I-II**) and outcome reported via the most recent QPRF (e.g., v 2.0 in March 2025) (10).
- Outcome of the DA application (hazard identification and potency categorisation according to UN GHS categories, or inconclusive result).
- Any deviation from the ITS DA.
- Conclusion.

3.2 References

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Part III SECTION 4 - Defined Approaches for Derivation of a Quantitative Point of-Departure

86. Part III of the Guideline includes Defined Approaches that allow for the derivation of a quantitative point of departure (PoD) for use in risk assessment. Additional detailed information can be found in the *Supporting document to the Guideline (GL) on Defined Approaches (DAs) for Skin Sensitisation* (1).

4.1 “SARA-ICE” Defined Approach for derivation of a point of departure

4.1.1. Summary

87. The Skin Allergy Risk Assessment-Integrated Chemical Environment (SARA-ICE) DA is a Bayesian statistical model which estimates an ED_{01} , which is provided as a probability distribution. The ED_{01} is an estimate of a dermal dose ($\mu\text{g cm}^{-2}$) at which there is a 1% chance of inducing sensitization from a human predictive patch test (HPPT). The SARA-ICE DA uses a set of defined input data and specifies a single point, computed as the geometric mean of the ED_{01} distribution conditional on it being less than $60,000 \mu\text{g cm}^{-2}$, which provides a point of departure (PoD) that can be applied within a quantitative risk assessment for skin sensitisation.

88. The SARA-ICE DA is based on data from test methods from KEs 1-4 of the adverse outcome in the AOP. The SARA-ICE DA is based on a previously published SARA model which initially included 81 chemicals in the database and was subsequently expanded to 434 chemicals, leveraging chemicals contained within the Integrated Chemical Environment (ICE, <https://ice.ntp.niehs.nih.gov/>, (2,3)). Further description of the database on which SARA-ICE is trained is provided in **Annex 10** of *Supporting document to the Guideline (GL) on Defined Approaches (DAs) for Skin Sensitisation* (1). Evaluation of SARA-ICE is described in paragraph 100 and performance is described in section 4.1.4. Test methods which can be used as input information sources include: the Direct Peptide Reactivity Assay (DPRA; OECD TG 442C), kinetic DPRA (kDPRA; OECD TG 442C), KeratinoSens (OECD TG 442D), the human cell line activation test (h-CLAT; OECD TG 442E), and U-SENS (OECD TG 442E), as well as *in vivo* methods such as the local lymph node assay (LLNA; OECD TG 429, OECD TG 442A, OECD TG 442B) and the human predictive patch test (HPPT) methods i.e., human repeat insult patch tests (HRIPT) and human maximization tests (HMT) (4–10). Data in any combination from these test methods can also be used to predict the ED_{01} . Note that the SARA-ICE model may accept existing LLNA data and HPPT data as inputs; however, new *in vivo* data should not be generated. The SARA-ICE model should be run using a minimum of two *in chemico* and/or *in vitro* information sources based on different KEs.

89. The SARA-ICE DA was evaluated first using posterior predictive checks, which are recommended for any Bayesian statistical model. Checks included a comparison of prior marginal distributions against posterior marginals to ensure model parameters are identifiable from the data. The SARA-ICE DA uses latent variables to represent average quantities for each input type. Checks that latent variables are both correlated with the ED_{01} and their respective input data in the SARA-ICE database were performed. Generating ED_{01} estimates for chemicals not within the core SARA-ICE database requires the joint posterior for the full database be approximated analytically. This approximation is then used as a prior within a SARA-ICE production DA to generate ED_{01} estimates for new input data. Further checks were performed to ensure the production version returns the correct estimates. Details of these analyses can be found in **Annex 11** of *Supporting document to the Guideline (GL) on Defined Approaches (DAs) for Skin Sensitisation* (1). The database on which the model was evaluated is static (finalized December 2022) and in case of subsequent version changes, it will be evaluated per OECD procedures.

4.1.2. Data Interpretation Procedure

90. The SARA-ICE DA uses a Bayesian statistical model to estimate the ED_{01} . A PoD is computed as the geometric mean of the ED_{01} distribution conditional on it being less than $60,000 \mu\text{g cm}^{-2}$. Technically,

the model can be run with no chemical-specific inputs, in which case the resulting ED₀₁ estimates will be taken directly from the prior distribution. Using chemical-specific data to inform estimates will, for any typical input, return an ED₀₁ estimate whose variance (uncertainty) is reduced relative to the prior. The magnitude of reduction in uncertainty is a function of several factors including input type(s), input numbers, and input concordance. To ensure an appreciable level of uncertainty reduction is achieved, a **minimum of 2 test methods per compound**, covering 2 separate KEs from within OECD TG 442C, 442D, or 442E, should be entered into the input template used to run the model ([LINK TO SARA-ICE APP](#)¹⁶). Admissible data types are indicated in **Table 4.1** and Paragraph 88. There is no restriction to the maximum number of stated data inputs, in type or experimental repeats, the model can include. Any available *in vivo* data may also be included, but new *in vivo* data should not be produced to be used in this DA. **Figure 4.1** demonstrates the workflow for use of SARA-ICE, and further details are provided in **Appendix III** of this document.

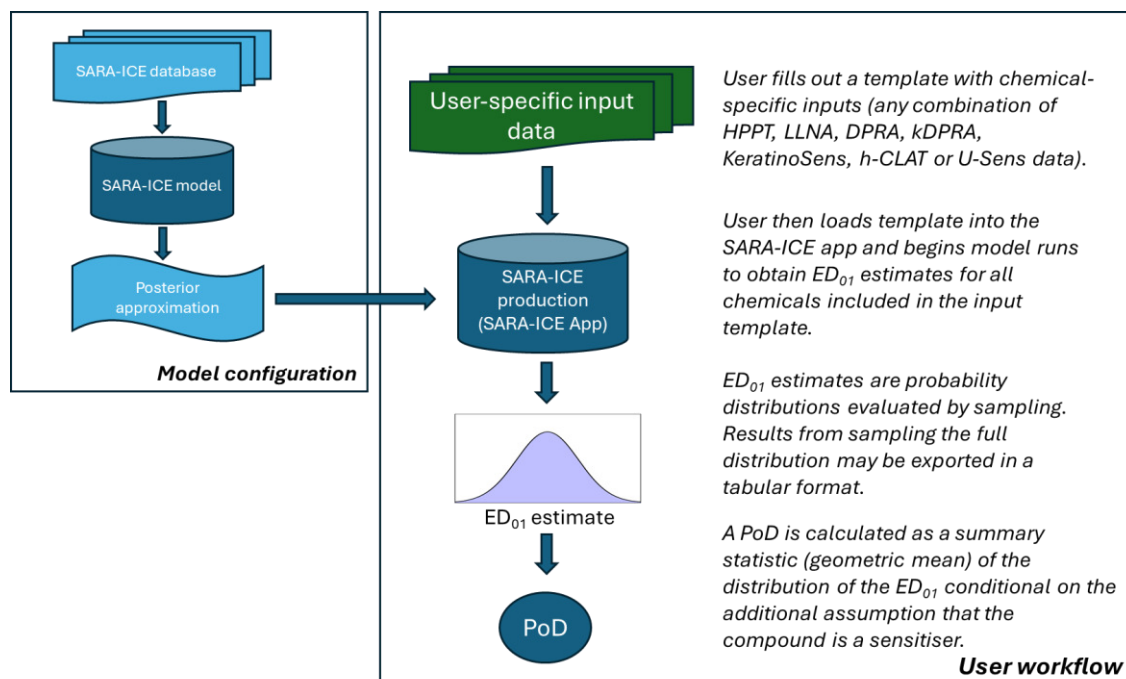
Table 4.1. Summary of Acceptable Test Methods for SARA-ICE

Test Methods	Test Guideline	Key Event	Data Input
DPRA	TG 442C	1	Percent depletion (Cys and Lys)
kDPRA	TG 442C	1	log kmax (units M ⁻¹ s ⁻¹)
KeratinoSens	TG 442D	2	EC _{1.5} and IC ₅₀ (μM)
h-CLAT	TG 442E	3	EC200 (CD54) EC150 (CD85) CV75 (μg mL ⁻¹)
U-SENS	TG 442 E	3	EC150 and CV70 (μg mL ⁻¹)
LLNA	TG 429, TG 442A, TG 442B	4	EC3 (percent)
HPPT	N/A	AO*	Dermal dose (μg cm ⁻² , N test subjects, N subjects sensitised)

*Adverse Outcome

¹⁶ <https://ntp.niehs.nih.gov/whatwestudy/niceatm/test-method-evaluations/skin-sens/da/SARA-ICE>

Figure 4.1. Workflow for the SARA-ICE DA



91. The model estimates the ED₀₁ as a probability distribution which is typically summarized as a 90% credible interval (the interval between the 5th and 95th percentiles of the distribution). The variance of the distribution is a measure of the uncertainty in the estimate and its width is dependent on the provided input data, such as the number, type, and concordance between inputs. For the purposes of defining a PoD for use in quantitative risk assessment for skin sensitisation, this distribution is truncated at 60,000 µg cm⁻² which is the maximum possible dose for a standard HRIPT. A PoD (geometric mean of the ED₀₁ distribution) is then calculated as a summary statistic of the distribution. The SARA-ICE Technical Specification in **Annex 10** of the *Supporting Document to the Guideline (GL) on Defined Approaches (DAs) for Skin Sensitisation* [1] provides further detail on the SARA-ICE PoD.

4.1.3. Description and limitations of the predictive model and its information sources

92. The individual *in chemico*, *in vitro*, and *in vivo* information is generated using existing OECD test guidelines (OECD TG 429, 442A, 442B, 442C, 442D, 442E [4-10]) and the protocols detailed therein. In the case of the kDPRA and KeratinoSens assays, commonly reported units are expressed as molar concentrations. If these are provided as inputs, the model also requires the specification of the chemical's molecular weight to enable conversion to mass concentrations. Note that the SARA-ICE model may accept existing LLNA data and HPPT data as inputs; however, new *in vivo* data should not be generated.

93. The following assays have been characterized and included as potential input information sources in the SARA-ICE DA:

- Direct Peptide Reactivity Assay (DPRA; OECD TG 442C; KE1 [7]): Skin sensitisers are generally electrophilic and react with the nucleophilic moieties of proteins. The DPRA measures depletion of two peptides containing either cysteine or lysine residues due to covalent binding. A test chemical that induces mean peptide depletion of cysteine- and lysine-containing peptide above 6.38% (or in the case of co-elution, cysteine-only depletion above 13.89%) is considered to be positive. The SARA-ICE DA does not make use of this classification rule. Instead, the input into SARA-ICE is the percentage depletion of both peptides. The SARA-ICE DA then determines the maximum

depletion of either of the peptides. Depletions less than 1% are interpreted as < 1%, whilst depletions greater than 99% are interpreted as > 99%. It is noted that while the DPRA gravimetric method has not been specifically evaluated for use in the SARA-ICE DA, given that the DPRA gravimetric evaluation that was conducted under OECD TG 442C found that the approaches are equivalent, it is expected to provide data within an acceptable range for use as an input into SARA-ICE.

- **Kinetic Direct Peptide Reactivity Assay (kDPRA; OECD TG 442C; KE1 [7]):** The kDPRA is intended to quantify the reactivity of test chemicals in a time and concentration dependent manner. The rate of depletion of the cysteine peptide (used within the DPRA) is determined. Kinetic rate constants are calculated and reported as the log Kmax value with Kmax having units $M^{-1} s^{-1}$. The kDPRA is used in the discrimination of UN GHS subcategory 1A from those not categorized as sub-category 1A. For SARA-ICE, the log Kmax is converted to units $g L^{-1} s^{-1}$ and multiplied by -1 (such that higher potency chemicals are associated with lower values). This is handled by the model, assuming the molecular weight of the chemical has been specified. If the reactivity rate is too slow to measure, then the input should be < X where X is the limit of detection of the test. For training purposes, this has been assumed to be $-3.5 M^{-1} s^{-1}$, thus, for chemicals exhibiting negligible reactivity in the kDPRA, the input should be < -3.5.
- **KeratiNoSens: (In vitro Skin Sensitisation: ARE-Nrf2 Luciferase Test Method; OECD TG 442D; KE2 [6]):** Keratinocytes harbouring a reporter gene construct react to possible sensitisers via the Nrf2-Keap1 pathway. A test chemical that causes > 1.5 fold luciferase induction, at viabilities > 70% when compared to the vehicle control, is considered to be positive. The SARA-ICE input is the $EC_{1.5}$ (μM) and the IC_{50} (μM). These are converted by the template to units of $\mu g mL^{-1}$, assuming molecular weight has been provided. The model also accepts inputs as > X or < X where X is a test concentration within the assay. This is required, for example, to input a negative test into the model (for which no $EC_{1.5}$ is defined). If the test is negative, then no luciferase induction occurs at any test concentration and the input into the model should > X where X is the maximum tested concentration. Similarly, if > 1.5 fold induction is observed at all test concentrations (i.e., the test is positive at all test concentrations), the input into SARA-ICE should be provided as < X where X is the minimum test concentration.
- **Human Cell Line Activation Test (h-CLAT; OECD TG442E; KE3 [4]):** Activation of antigen presenting cells is characterized by the up-regulation of CD86 and/or CD54. The h-CLAT is considered to be positive if CD86 induction exceeds 1.5 fold and/or CD54 exceeds 2 fold at viabilities > 50% when compared to the vehicle control. The SARA-ICE input is both the CD54 EC_{200} and the CD86 EC_{150} ($\mu g mL^{-1}$) and the CV75 ($\mu g mL^{-1}$). If either value is not reported because no change in the biomarker was observed, the input is > X where X is the maximum concentration tested. In the case that induction exceeds thresholds at all test concentrations for either biomarker (i.e., the test is positive for that biomarker at all test concentrations), the model may also accept < X where X is the minimum test concentration.
- **U-SENS (OECD TG 442E; KE3 [4]):** Activation of antigen presenting cells is characterised by the up-regulation of CD86. The U-SENS is considered positive if the CD86 induction exceeds 1.5-fold at viabilities > 70% compared to vehicle control. The SARA-ICE input is the CD86 EC_{150} ($\mu g mL^{-1}$) and CV70 ($\mu g mL^{-1}$). If the test is negative, the input to the model is > X where X is the maximum concentration should be used. In the event that CD86 induction exceeds the 1.5-fold at all test concentrations, the input should be < X where X is the minimum test concentration.
- **Local Lymph Node Assay (LLNA; OECD TG429, 442A, 442B; KE4 [5,8,9]):** The induction phase of skin sensitization (KE4) is characterized by lymphocyte proliferation in the lymph nodes draining the site of the exposure and is proportional to the dose and potency of the substance. The LLNA

measures proliferation relative to the vehicle control treated group for mice exposed on their ear to a test substance of interest. The ratio of the mean proliferation for each treatment group compared to the vehicle control is the Stimulation Index (SI). An $SI \geq 3$ is considered to be indicative of skin sensitization. The input for SARA-ICE template is the EC3 expressed as a percent. These are converted to $\mu\text{g cm}^{-2}$ by the model. The model may also accept an input for negative tests expressed as $> X$ where X is the maximum tested concentration. Alternatively, if all test concentrations result in an SI greater than 3, the recommended input into SARA-ICE is $< X$ where X is the lowest concentration tested. This also allows single-dose LLNA studies to be input in the SARA-ICE DA. If a single-dose LLNA is run at concentration X and the $SI < 3$, then the input into SARA-ICE is $> X$. Conversely, if the $SI > 3$, the study result can be entered as $< X$. Only existing LLNA data are recommended for use in the SARA-ICE model.

- Human Predictive Patch Test (HPPT [10]): HPPT methods include the human maximization test (HMT) and the human repeat insult patch test (HRIPT). Both test methods quantify an empirical sensitization rate following exposure to a (potential) allergen at some chosen dermal dose. The HMT incorporates a pre-treatment with an irritant and typically has a smaller test pool than HRIPT. Required HPPT inputs into the SARA-ICE DA are the dermal dose expressed as $\mu\text{g cm}^{-2}$, the total number of test subjects and the total number of subjects sensitized following the challenge.

94. The current limitations of the individual test methods, such as with respect to solubility, are described in the respective test guidelines and the validation studies cited therein. No filtering of assay data was performed prior to training the SARA-ICE DA, therefore uncertainty measures provided by the DA should reflect such limitations. However, caution is advised when selecting inputs since biases may nevertheless occur (e.g., when using a DPRA input for a chemical whose sensitization potential is enhanced via metabolic activation). Only valid assay results, i.e. results within the applicability domain of each assay used as input, should be used to run the DA for regulatory purposes. Evaluation of data quality should be made prior to use in the DA, this includes use of human data for which Strickland et al., 2023 [10] can be used for reference.

4.1.4. Confidence in the SARA-ICE DA predictions

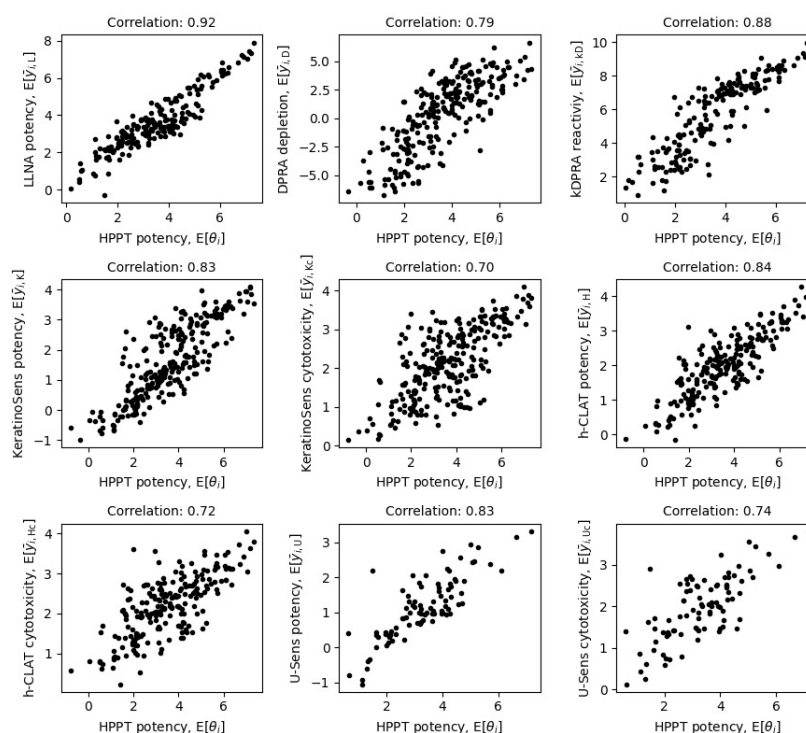
95. The SARA-ICE DA is built upon a Bayesian statistical model and as such provides potency estimates with a numerical measure of precision. The provided distribution represents the range of ED_{01} values that are both consistent with user-specified input data and the model – where the model is a characterization of the association between ED_{01} estimates and assay data for a training set of 434 chemicals. Further information on the SARA-ICE database can be found in the Technical Specification in Appendix 2. A decision on whether to use specific source data is dictated by the limitations of the different input methods found within the Annex of the requisite test method (TG 429, TG 442A, TG 442B, TG 442C, TG 442D, TG 442E). As indicated, for a higher confidence estimate, at least 2 methods that cover 2 key events should be used as input data. A greater number of data inputs will likely also increase confidence (e.g. decrease the width of the distribution) of the model ED_{01} output depending on the concordance of the additional data utilized [3]. All available high-quality data is encouraged to be used.

96. A number of basic model checks were carried out to ensure that the SARA-ICE model is behaving as expected. Posterior predictive checks show that the marginal posterior distribution of each parameter is not concentrating in the extreme tail of the prior, indicating that the chosen prior is only weakly informative on the posterior estimate. Therefore, estimates of model parameters would be unlikely to change if prior choices were relaxed. Further checks were completed to confirm the model assumption that there is a linear relationship with non-zero slope to the predictors included in the model. Figure 4.2 shows that each of the possible inputs is informative towards estimation of the ED_{01} . Further analysis shows that location-like latent variables for each predictor are both correlated with the ED_{01} and experimental repeats for

chemical. The SARA-ICE Technical Specification in **Annex 10** of the *Supporting Document to the Guideline (GL) on Defined Approaches (DAs) for Skin Sensitisation* [1] provides a detailed explanation of the basic model checks carried out.

Figure 4.2. Correlations between the SARA-ICE representation of sensitiser potency (abscissa) versus SARA-ICE representation of the average assay result (ordinate), after transformation.

Each point corresponds to a distinct compound within the SARA-ICE database provided that the compound has both HPPT and assay-specific data available.



4.1.5. Proficiency of SARA-ICE DA

97. The SARA-ICE DA uses a statistical interpretation procedure, and no expert judgement is required to derive the PoD and its probability distribution using the model. Proficiency chemicals for the individual *in chemico* and *in vitro* information sources (KE1, KE2, KE3) are defined in the respective guidelines (OECD TG 442C, 442D, 442E) ([2,4,5]). The protocol details for the SARA-ICE DA are included in **Appendix III** of this guideline. Proficiency has been demonstrated for the SARA-ICE DA v1, and this is the software version that is intended for use with this DA. Proficiency for the individual information sources demonstrates proficiency for the DA.

4.1.6. Predictive Capacity of the SARA-ICE DA

98. SARA-ICE derives a point of departure for risk assessment. The model estimates the dose at which skin sensitisation is likely to occur in 1% of an HPPT-eligible human population (ED_{01}), reported as a distribution with associated confidence intervals. For PoD estimation, an upper bound is imposed on the ED_{01} distribution at a threshold of $60,000 \mu\text{g cm}^{-2}$. This reflects the bound for a GHS classification of NC and encodes an implicit assumption that the chemical under consideration for PoD based risk assessment

is a sensitiser. The PoD is therefore defined as the geometric mean of the distribution of the ED₀₁ conditional on the ED₀₁ being less than 60,000 µg cm⁻².

99. PoDs were computed for all 196 chemicals in the DASS evaluation set (Annex 2) using a) *in chemico* and *in vitro* data also reported in the evaluation set (which consists of a single input for DPRA, KeratinoSens and h-CLAT) and b) a single kDPRA sourced from Natsch et al., 2022 [12], if available). It should be noted that 187 / 196 (95%) of chemicals in the DASS evaluation set are presented in the SARA-ICE database. However, the data used for training versus evaluation differs for most chemicals. For training purposes, a chemical may have both *in vivo* and *in vitro* data with multiple studies per data type. For evaluation, inputs are restricted to a single DPRA, single KeratinoSens, single h-CLAT and, if available, a single kDPRA study. It follows that ED₀₁ estimates obtained during training may be considerably different from those obtained in the evaluation.

100. SARA-ICE PoDs, were compared to reference DSA1+ values reported in Herzler et al., 2024 [11]. The reference value is termed the *median sensitization potency estimate* (MSPE) and is reported as weight-of-evidence potency metric calculated from human data. A DSA1+ MSPE value was defined for 54 / 196 chemicals in the DASS evaluation set. A positive correlation exists (Pearson correlation coefficient is 0.64) and 46 / 54 (85%) of SARA-ICE PoDs are within 10-fold of the reference value. Roughly one third, 19 / 54 (35%) of SARA-ICE PoDs are within 2-fold of the reference value. The same SARA-ICE PoDs were also compared to 154 chemicals with reference LLNA MLLPs sourced from OECD DASS reference list [1]. A positive correlation exists (Pearson correlation coefficient is 0.71) and 122 / 154 (79%) of SARA-ICE PoDs are within 10-fold of the reference value. Again, roughly one third, 49 / 154 (32%) of SARA-ICE PoDs are within 2-fold of the reference value. SARA-ICE PoDs exhibit almost no bias relative to human reference values (mean and median fold-differences are 1.1). Relative to LLNA reference values, the SARA-ICE PoD is nearly 2-fold lower, on average. These comparisons are presented in **Figure 1.1**, section 1.2. Further analysis was conducted based upon fewer inputs, no appreciable difference in prediction bias was observed for the minimum input requirement of two separate key event-based model inputs. Further analyses are made in **Annex 11** of the Supporting document to the Guideline (GL) on Defined Approaches (DAs) for Skin Sensitisation (1).

4.1.7. Reporting of the DA

101. The reporting of the SARA-ICE DA should follow the template described in OECD GD 255 [13], and should include at a minimum the following elements:

- Test chemical identification (e.g. chemical name, structural formula, composition, isomers, impurities including their quantities as available, CAS number, batch and lot number, and other relevant identifiers)
- Individual test reports including raw data for the individual tests performed per corresponding guideline (OECD TG 429, 442A, 442B, 442C, 442D, 442E, or the HPPT methods, Strickland et al., 2023 [10]). Note that the chemical identity for each test report should match that above.
- The input template containing individual assay datapoints used for computation in SARA-ICE.
- Outcome of the DA application (PoD (geometric mean and 90th credible interval of the distribution of the ED₀₁ conditional on the ED₀₁ being less than 60,000 µg cm⁻²))
- Any deviation from the SARA-ICE DA
- Conclusion

4.1.8. References

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4.2 “Regression-based” Defined Approach for derivation of a point of departure for skin sensitization

4.2.1 Summary

102. The regression-based defined approach (DA) is based on multiple linear regression models. As input data, quantitative (dose-response and kinetic) data from *in vitro* and *in chemico* assays addressing KE1, KE2 and KE3 of the AOP are used. Different models were trained on (i) EC3 values from the Local Lymph Node Assay (LLNA) [1, 2] and (ii) on weight-of-evidence (WoE) Potency Values (PV)[3] derived from human DSA04 and NOEL values and LLNA EC3 values [4]. These individual regression models are combined in the DA to provide a finite point of departure (PoD) that can be applied within a quantitative risk assessment for skin sensitisation [5]. The PoD is both expressed as a value in % in a vehicle applied at 25 mg/cm² to the skin (similar to the LLNA) or as a dose per unit metric in µg/cm². The conversion factor between these two metrics is a fixed factor of 250.

103. The details of the training of the regression models and the set of chemicals used are described in detail in the open access literature [1-3]. The database with all the input data and the prediction of the PoD for all chemicals tested (n = 220) is provided in **Annex 12.a** of Supporting Document to this document. The specific test methods in the KE-based test guidelines which can be used as input information sources include: the kinetic DPRA (kDPRA; OECD TG 442C), KeratinoSens (KS; OECD TG 442D) and the human cell line activation test (h-CLAT; OECD TG 442E) [6-8]. Data from at least two of these information sources are required to predict the PoD. For chemicals, which are rated as sensitisers according to a DA described in Part I or Part II of this guideline, the input data shall include quantitative results from at least two positive KE outcomes, whereby the kDPRA is conducted as a quantitative test for KE1 in case of a positive DPRA. For chemicals positive in DPRA and KS, data from at least kDPRA and KS shall be used. For chemicals positive in DPRA and h-CLAT, data from at least kDPRA and h-CLAT shall be used. For chemicals positive in KS and h-CLAT, at least these data shall be used. For chemicals tested in all three KE tests and positive in at least two tests, all available evidence including the kDPRA result shall be used, see Figure 4.3. In addition, vapour pressure is used to correct for volatility under topical application. Evaluation of predictivity of the DA is described in section 4.2.6.

4.2.2 Data Interpretation Procedure

104. The regression models use logarithmically transformed and normalised data from the individual information sources and the input parameters as summarised in Table 4.2. The models combine these data in linear regression equations with coefficients trained on the full dataset. These calculations are made in a fully transparent spreadsheet which is provided as **Annex 12.b** of the Supporting Document to this guideline [11]. The measured parameters summarised in Table 4.2 can be directly entered into the spreadsheet to arrive at a conclusion. Depending on the available input data, multiple predictions are made: (i) Equations EQ1, EQ4, EQ5 and EQ6 were trained against LLNA data (see[1]), while (ii) EQ1d, EQ4d, EQ6d and EQ5e were trained against the PV (see [3]). The DA then automatically selects, depending on the KE covered by the input data, a PoD according to the workflow in Figure 4.3 below, taking the more conservative estimate from the two models trained on either LLNA EC3 or PV values.

Figure 4.3. Use of the published predictive equations based on different combinations of input data.

The DA uses the appropriate model depending on which tests are available. A conservative choice is made, taking the lower of the two values from either (i) the models trained on LLNA EC3 values (EQ1, EQ4, EQ6 and EQ5; see [1]) or (ii) trained on the PV values (EQ1d, EQ4d, EQ6d and EQ5e; see [3]).

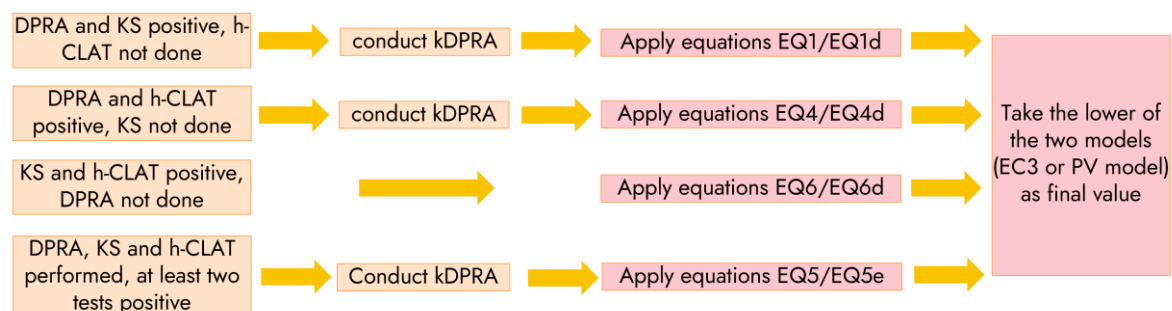


Table 4.2. Summary of Test Methods and input parameters for the regression-based DA

Test Methods	Test Guideline	Key Event	Data Input and unit
kDPRA	TG 442C	1	$\log k_{\max}$ ($M^{-1} s^{-1}$)
KeratinoSens	TG 442D	2	EC _{1.5} and IC ₅₀ (μM ; $\mu g mL^{-1}$ for chemicals with no defined molecular weight)
h-CLAT	TG 442E	3	EC200 (CD54), EC150 (CD86) and CV75 ($\mu g mL^{-1}$)
Vapor pressure	Vapor pressure in Pascal (Pa) as calculated e.g. with the TIMES SS model or e.g. on https://comptox.epa.gov/dashboard/ , the later is in mm Hg, mmHg can be transformed to Pa by multiplying with 133		

105. The regression-based DA provides a finite value. This PoD is a ‘most likely’ approximation of the LLNA EC3 value or the human DSA04 value, i.e. the concentration which in a human repeat insult patch test (HRIPT) is expected to sensitize 4% of the tested individuals. LLNA EC3 expressed in $\mu g/cm^2$ and human DSA04 in $\mu g/cm^2$ are well correlated in terms of absolute potency, while human DSA1+ and human DSA01 tend to be lower than the LLNA EC3 values (see **Annex 12.a** of the Supporting Document to this guideline [11]). As the regression-based DA is trained both on human and LLNA data, the human metric best correlated with the LLNA metric was chosen. Both these human data and the LLNA EC3 value represent the lower end of the sensitization induction dose-response curve. Appropriate safety factors, depending on the use situation, shall be applied to this PoD in quantitative risk assessment, which is not within the scope of this guideline.

4.2.3 Description and limitations of the predictive model and its information sources

106. The individual *in chemico*, *in vitro*, and *in vivo* information is generated using existing OECD test guidelines (OECD TG 442C, 442D, 442E [6-8].) and the protocols detailed therein. In the case of the kDPRA and the KeratinoSens assays, commonly reported units are expressed as molar concentrations. The model requires the specification of the chemical's molecular weight to enable conversion to mass concentrations. In case the molecular weight is unknown (mixtures and multi-constituent substances of unknown composition (i.e. UVCB substances of unknown or variable composition, complex reaction products or biological materials), a default molecular weight of 200 g/mol can be used as applied in the gravimetric DPRA (442C). This default molecular weight is also used in the gravimetric KeratinoSens prediction model (442D). The regression-based DA can be run using data from a minimum of two KE-based *in chemico* and/or *in vitro* information sources applicable to a particular chemical. Applicability

domain limitations as outlined in TG442c, 442d and 442e for a particular chemical may guide which assay is being conducted. If a chemical is rated as a sensitizer due to at least two positive test results in TG 442C, 442D, 442E, at least the data from two positive KE shall be used whereby the kDPRA is conducted as a quantitative test for KE1 in case of a positive DPRA. For chemicals positive in DPRA and KS, data from at least kDPRA and KS shall be used. For chemicals positive in DPRA and h-CLAT, data from at least kDPRA and h-CLAT shall be used. For chemicals positive in KS and h-CLAT, at least these data shall be used. For chemicals tested in all three KE tests and positive in at least two tests, all available evidence including the kDPRA result shall be used, see Figure 4.3. For chemicals for which the kDPRA is not applicable, potency is assessed by a combination of KS and h-CLAT. For details on the underlying rationale and for possible testing schemes see [2]. All the test methods used for the regression DA are technically applicable to test mixtures and considerations of the different test methods when using a gravimetric approach should be followed. When assessing mixtures and no experimental vapor pressure is known, the vapor pressure is set to zero. This is a conservative approach as the actual vapor pressure can only reduce the predicted sensitization potency. The models have been trained on the three specific KE-based tests and using large sets of dose-response data. If for other KE-based tests such quantitative datasets are available, similar regression-based models may be derived in the future in analogy to [1, 2]. However, these would need completely independent validation and assessment for predictivity and the development of an independent calculation worksheet. Therefore, currently the DA is only applicable to the prescribed methods.

107. The following assays have been characterized and included as input information sources in the regression-based DA:

- Kinetic Direct Peptide Reactivity Assay (kDPRA; OECD TG 442C; KE1 [7]): The kDPRA is intended to quantify the reactivity of test chemicals in a time and concentration dependent manner. The rate of depletion of the cysteine peptide (used within the DPRA) is determined. Kinetic rate constants are calculated and reported as the Log $k_{\max}$ value with $k_{\max}$ having units $M^{-1} s^{-1}$. For chemicals with no measurable reactivity above the threshold in the kDPRA, the limit value of the Log $k_{\max} = -3.5$ is entered into the model as specified in the Spreadsheet in **Annex 12.b** of Supporting Document to this guideline [11]. For chemicals with a high reactivity at the earliest time point measured (10 min in the standard setting of the kDPRA), testing at shorter incubation times should be considered in order to not underestimate potency (as indicated in TG442C, ANNEX III, para 18).
- KeratinoSens: (In vitro Skin Sensitisation: ARE-Nrf2 Luciferase Test Method; OECD TG 442D; KE2[6]): Keratinocytes harbouring a reporter gene construct react to possible sensitizers via the Nrf2-Keap1 pathway. A test chemical that causes > 1.5-fold luciferase induction, at viabilities > 70% when compared to the vehicle control, is considered to be positive. The model input is the $EC_{1.5}$ (μM) and the IC_{50} (μM). For chemicals with no defined molecular weight, these can be entered in $\mu g mL^{-1}$. If no luciferase induction is observed at any test concentration (i.e., the test is negative), the input into the model should be 4000 μM (or 800 $\mu g mL^{-1}$), i.e. twice the maximal concentration tested. The same approach applies to the IC_{50} value in case no cytotoxicity is observed as specified in the Spreadsheet in **Annex 12.b** of Supporting Document to this guideline [11].
- Human Cell Line Activation Test (h-CLAT; OECD TG442E; KE3 [8]): Activation of antigen presenting cells is characterized by the up-regulation of CD86 and/or CD54. The h-CLAT is considered to be positive if CD86 induction exceeds 1.5-fold and/or CD54 exceeds 2-fold at viabilities > 50% when compared to the vehicle control. The model input is both the CD54 EC_{200} and the CD86 EC_{150} ($\mu g mL^{-1}$) and the CV75 ($\mu g mL^{-1}$). If either value is not reported because no change in the biomarker was observed or cytotoxicity is observed at the maximal test concentration, the input is 5000 $\mu g mL^{-1}$.

108. If multiple test results are available for a given test, they are assessed for their validity based on the criteria laid out in the KE-based test guidelines. If multiple valid and concordant results are available, the geometric mean of the individual parameters from multiple tests is used in the calculation spreadsheet. In cases where multiple positive and negative results are available for a given KE-event based test, the geometric mean of the data from the positive tests is used.

109. The current limitations of individual *in chemico* and *in vitro* test methods, such as limitations with respect to solubility, are described in the respective test guidelines (TG 442C, Appendix 1; TG 442D, Appendix 1A; TG 442E, Annex 1) and the validation studies cited therein (2, 3, 4). No filtering of assay data to exclude chemicals outside of the application domain was performed prior to training the regression models, therefore uncertainty measures provided by the DA should reflect such limitations. Specifically, (i) data for chemicals falling into the borderline range of the individual methods were still used and (ii) h-CLAT data were used independently of the cLogP of the test chemical when the models were trained. However, caution is advised when selecting inputs since biases may nevertheless occur (e.g., when using a kDPRA input for a chemical whose sensitisation potential is enhanced via metabolic or abiotic activation, or for using the kDPRA for chemicals which are predominantly reacting with amine groups as shown e.g. by the reaction with the Lysine peptide in the DPRA; limitations of the kDPRA are summarised in TG442C, APPENDIX III, ANNEX 1). Therefore, only valid assay results, i.e. results within the applicability domain of each assay, should be used to run the DA for regulatory purposes. Evaluation of data quality should be made prior to use in the DA.

4.2.4 Confidence in the regression-based DA predictions

110. The regression equations were built upon a training set of $n = 139 - 203$ chemicals. Further information on the underlying database can be found in the **Annex 12.b** and in [1-3]. A decision on whether to use specific source data is dictated by the limitations of the different input methods found within the Annex of the respective test method (TG 442C, TG 442D and TG 442E). At least 2 methods that cover 2 key events should be used as input data.

111. The correlation of individual information sources and their combination to sensitiser potency in the LLNA has been analysed in the literature [9]. The stability of the regression models on different datasets have been analysed in detail, indicating a high robustness of the regression-based models [1, 2]. The consistency of the result when using three or only two information sources was investigated in supporting file ESM3 to [1], which is accessible under [10.14573/altex.2201141](https://doi.org/10.14573/altex.2201141). The performance of the individual regression models was shown as summary statistics in Table 2 in [1] and in Table 4 in [3], and for individual chemicals in the file ESM1 to [1] accessible under [10.14573/altex.2201141](https://doi.org/10.14573/altex.2201141). A high consistency of predictions was also observed if the models were trained on LLNA EC3 only or on PV, with a lower importance of cytotoxicity and a higher contribution of cellular activation markers for models trained on PV [3] (see Table 4 in [3]). The uncertainty in predicting *in vivo* data using these models should also be viewed in the context of the inherent variability of human and animal reference data and the correlation between these data, see e.g. [10].

4.2.5 Proficiency Chemicals

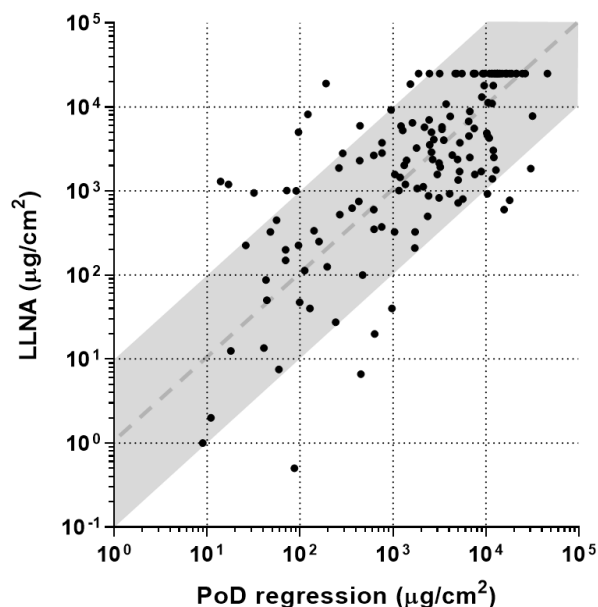
112. The regression models use a fixed data interpretation procedure to derive the PoD. Proficiency chemicals for the individual *in chemico* and *in vitro* information sources (KE1, KE2, KE3) are defined in the respective guidelines (OECD TG 442C, 442D, 442E) ([2,4,5]). The details to apply the regression models are directly included in the Spreadsheet, which forms **Annex 12.b** of the Supporting Document to this guideline. Proficiency for the individual information sources demonstrates proficiency for the DA.

4.2.6 Predictive Capacity of the regression-based DA

113. The regression-based DA derives a PoD for risk assessment. The model estimates the dose of the EC3 value in the LLNA or the dose at which skin sensitisation is likely to occur in 4% of a HPPT-eligible human population (DSA04 value). DSA04 value in $\mu\text{g}/\text{cm}^2$ are correlated to EC3 values expressed in $\mu\text{g}/\text{cm}^2$ and both values represent the lower end of the dose-response curve for experimental induction of skin sensitization. These values are combined in the PV [3, 4]. The PoD is therefore defined as a best estimate for either the LLNA EC3 or the WoE PV. PoDs were computed for 220 chemicals listed in Annex 4.2.A. The PoDs were compared to reference EC3 values or PV. When evaluating the PoD vs. the LLNA EC3 MLLP in the DASS database (n=146), 126/146 (86%) of regression-based PoDs are within 10-fold of the reference value, with 8/146 (5.5%) having a > 10-fold underpredicted potency. Furthermore, 59/146 (40%) of the PoDs are within 2-fold of the reference value. When the PoD are compared to the LLNA EC3 in the full database (n=220; including those EC3 values non-evaluated or excluded by the DASS review), a similar result is obtained: 191/220 (87%) of regression-based PoDs are within 10-fold of the reference value, with 13/220 (6.3%) having a > 10-fold underpredicted potency. Furthermore, 80/220 (36%) of the PoDs are within 2-fold of the reference value. The same PoDs were compared to the PV (n=139). 120/139 (86%) of PoDs are within 10-fold of the reference value, with 12/139 (8.9%) having a > 10-fold underpredicted potency. Furthermore, 43/139 (31%) of PoDs are within 2-fold of the reference value. Regression-based PoDs give a balanced prediction relative to the chosen reference values (geometric mean fold-differences vs. LLNA is 0.86 and vs. PV it is 0.99). Further analyses are made in **Annex 12.c** in the Supporting Document of this guideline [11]. Figure 4.4 shows regression-based PoDs compared to reference LLNA MLLPs in the DASS evaluation database.

Figure 4.4. All chemicals with in vitro data, compared versus OECD MLLP LLNA EC3 data (n=146).

The grey dashed line shows the line of identity, and the zone indicates regression DA based PoDs that are within 10-fold of the reference LLNA MLLP values.



To further compare predicted values vs. reference values, for each chemical the absolute fold difference is calculated according to the formula:

$$\text{Absolute fold difference} = 10^{|Log(\text{Predicted PoD}) - Log(\text{Reference value})|}$$

This gives the fold absolute deviation of the predicted PoD from the reference value in either direction.

In parallel for each chemical the fold difference is calculated according to the formula:

$$\text{Fold difference} = 10^{(Log(\text{Predicted PoD}) - Log(\text{Reference value}))}$$

A geometric mean of the fold difference close to 1 indicates, that the model is balanced and has no bias towards under- or over-prediction. A value below one indicates a model biased for conservative predictions and a value above one indicates an overall bias for underprediction.

The results on the individual chemicals are given in **Annex 12.a** of Supporting Document to this guideline [11] and a detailed summary analysis and discussion is presented in **Annex 12.c** of Supporting Document to this guideline [11] and summarised below in the form of the geometric mean and the median values:

Table 4.3. Summary of prediction accuracy on different datasets

	Geometric mean fold-difference	Geometric mean absolute fold-difference	Median absolute fold-difference
Full set (n=220) vs. LLNA EC3	0.86	3.46	2.56
Set with three tests (n=188) vs. LLNA EC3	0.88	3.57	2.63
Set with OECD MLLP (n=146) vs. LLNA EC3	0.86	3.50	2.47
Set with PV (n=139) vs. LLNA EC3	0.96	3.42	2.42
Set with PV (n=139) vs. PV	0.99	3.67	2.70

4.2.7 Reporting of the DA

114. The reporting of the regression-based DA should follow the template described in OECD GD 255 (6), and should include at a minimum the following elements:

- Test chemical identification (e.g. chemical name, structural formula, composition, isomers, impurities including their quantities as available, CAS number, batch and lot number, and other relevant identifiers)
- Individual test reports including raw data for the individual tests performed per corresponding guideline (OECD TG 442C, 442D, 442E). Note that the chemical identity for each test report should match that above.
- A copy of the prediction spreadsheet with input data and results
- Outcome of the DA application: Point of departure derived, indication on which regression equation and information sources the PoD is based on
- Assessment whether the chemical is in the applicability domain of the individual information sources
- Any deviation from the regression-based DA
- Conclusion

4.2.8 References

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10. Natsch, A., Griem, P., Irizar, A., Bridges, J., Vey, M., Lee, I., Api, A., Kern, P., Kimber, I., *Derivation of a Point of Departure using NAMs for application in Quantitative Risk Assessment of fragrance materials*. Regulatory Toxicology and Pharmacology, 2026: p. 106052.
10. OECD (2026), Supporting document to the Guideline (GL) on Defined Approaches (DAs) for Skin Sensitisation, Series on Testing and Assessment, No. 336, OECD Environment, Health and Safety, Paris. Available at: <https://www.oecd.org/en/topics/sub-issues/testing-of-chemicals/publications-on-testing-and-assessment-of-chemicals.html>

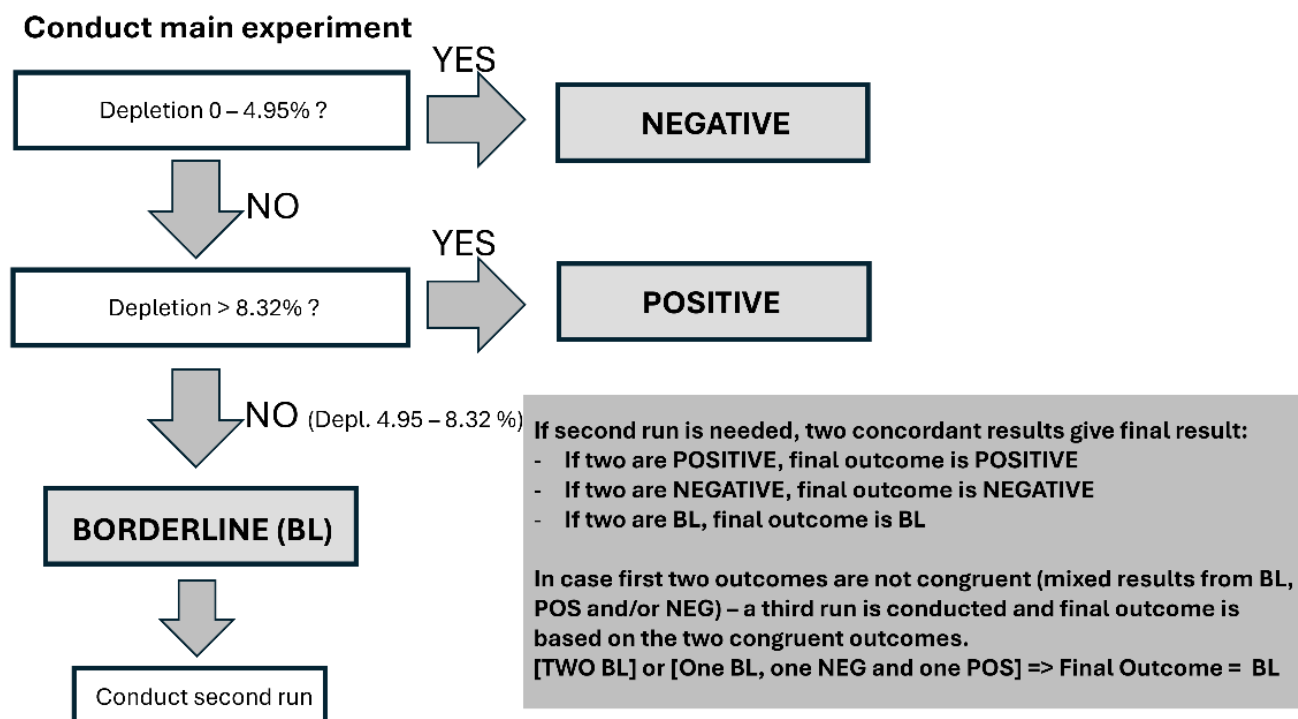
Annex 1. Prediction models for the individual *in chemico/in vitro* tests with multiple runs for use in 2o3 DA

115. Each of the test methods included in the 2o3 DA have different prediction models that may require multiple runs (independent repetitions) dependent upon the outcome of an individual run. An adaptation of the prediction model was used to determine borderline cases in the individual runs for the purpose of making predictions within the 2o3 DA. These adaptations (Figures A 1.1- A 1.8) below should be used in these methods to come to the final conclusion of the individual tests. Methods included in the 2o3 include: DPRA, ADRA, KeratinoSens™, LuSens, EpiSensA, h-CLAT, U-SENS™, IL-8 Luc assay, and GARD@skin. The methodologies to derive BRs for LuSens, ADRA, EpiSensA, IL-8 Luc and U-SENS™ are available in publications (1-4). BRs for GARD@skin was derived using same methodologies as found in (1). Further details for the derivation of the BRs for DPRA, EpiSensA, h-CLAT, and KeratinoSens are available in **Annex 7** of the *Supporting document to the Guideline (GL) on Defined Approaches (DAs) for Skin Sensitisation* (5).

DPRA Prediction Model

Figure A 1.1. Flow-chart of the DPRA prediction model (mean depletion) taking into borderline ranges and multiple runs conclude on borderline results within the 2o3 DA.

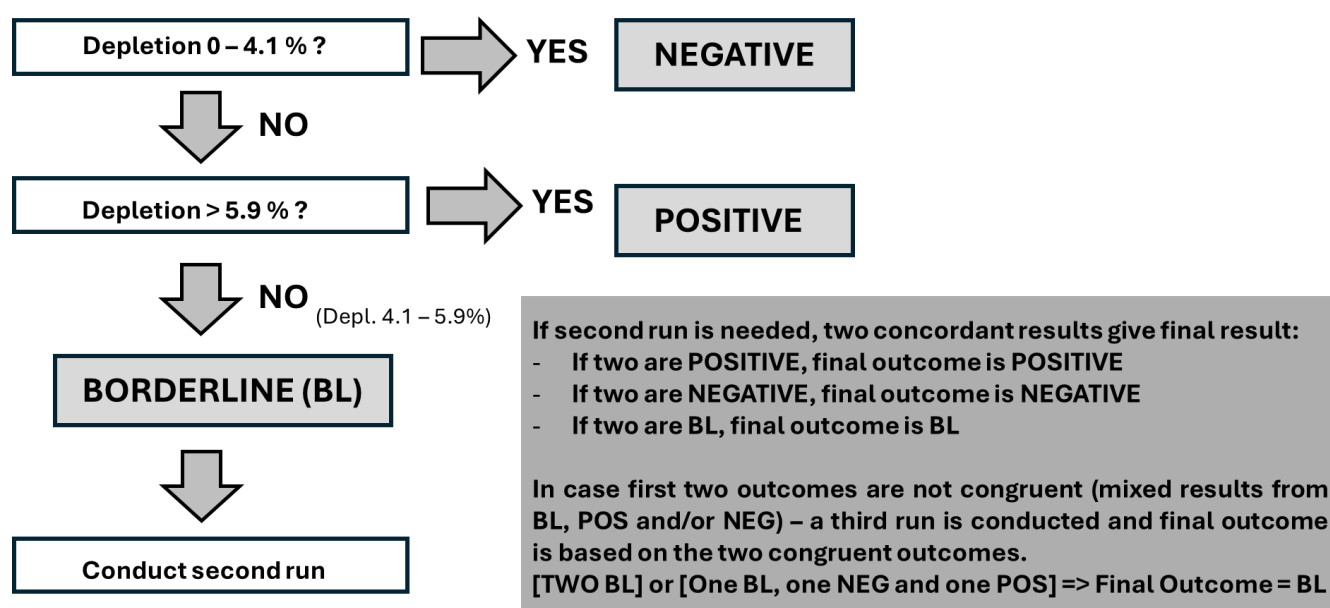
The original threshold for a positive classification is 6.38%, and the statistically derived borderline range around this threshold is 4.95% - 8.32%. The same flowchart applies to the cysteine-only prediction model, whereby the following thresholds apply: 10.56 % instead of 4.95% and > 18.47 % instead of >8.32%.



ADRA Prediction Model

Figure A 1.1. Flow-chart of the ADRA prediction model (mean depletion) taking into account borderline ranges and multiple runs to conclude on borderline results within the 2o3 DA.

The original threshold for a positive classification is a depletion value of 4.9% or greater for either probe, and the statistically derived borderline range around this threshold is 4.1% - 5.9%. The same flowchart applies to the NAC-only prediction model, whereby the following thresholds apply: 5.6% instead of 4.9%, 4.7% instead of 4.1%, 4.7% instead of 4.1% and > 6.8% instead of 5.9%.

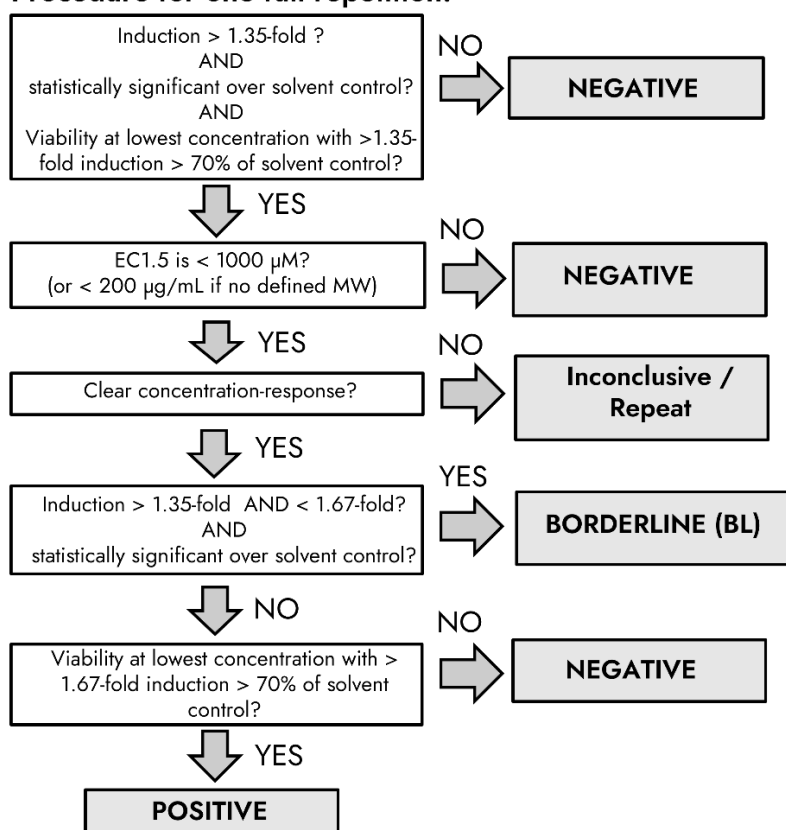


KeratinoSens Prediction Model

Figure A 1.1. Flow-chart of the KeratinoSensTM prediction model taking into account borderline ranges and multiple runs to conclude on borderline results within the 2o3 DA.

The original threshold for a positive classification is 1.5-fold induction, and the statistically derived borderline range around this threshold is 1.35 – 1.67-fold. Note: An independent run is referred to as ‘repetition’ in 442D, while it is called a ‘run’ in 442C and 442E; these nomenclatures do mean the same thing.

Procedure for one full repetition:



Perform at least two independent repetitions.

- If two are **POSITIVE**, final outcome is **POSITIVE**
- If two are **NEGATIVE**, final outcome is **NEGATIVE**
- If two are **BL**, final outcome is **BL**

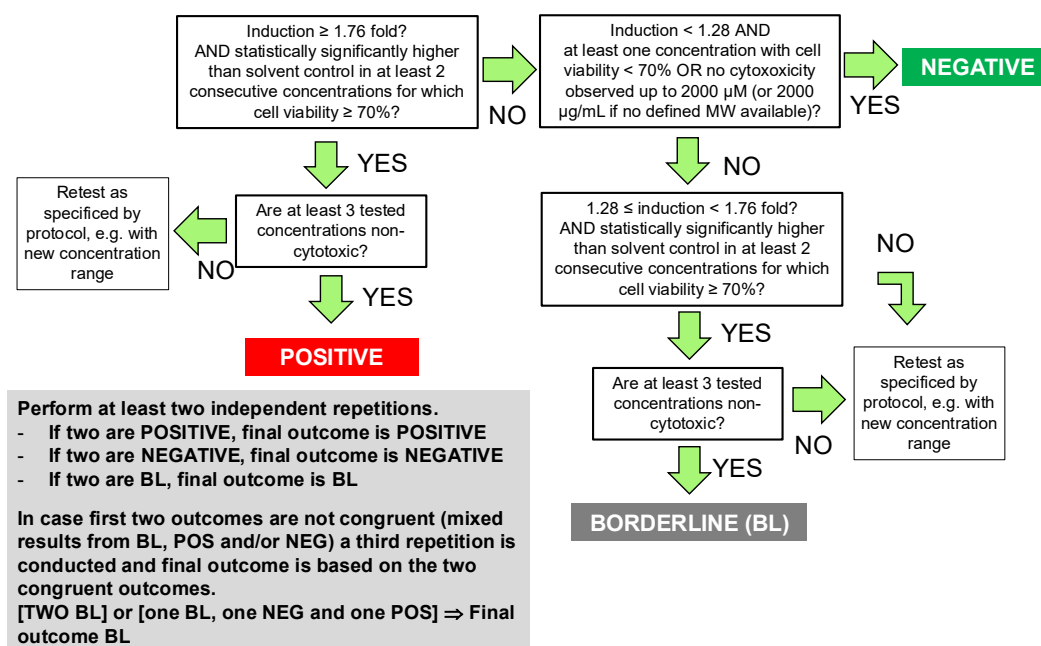
In case first two outcomes are not congruent (mixed results from BL, POS and/or NEG) a third repetition is conducted and final outcome is based on the two congruent outcomes. [TWO BL] or [one BL, one NEG and one POS] ⇒ Final outcome BL

LuSens Prediction Model

Figure A 1.1. Flow-chart of the LuSens prediction model, taking into account borderline ranges and multiple runs to conclude on borderline results within the 2o3 DA.

The original threshold for a positive classification is 1.5-fold induction, and the statistically derived borderline range around this threshold is 1.28 – 1.76-fold.

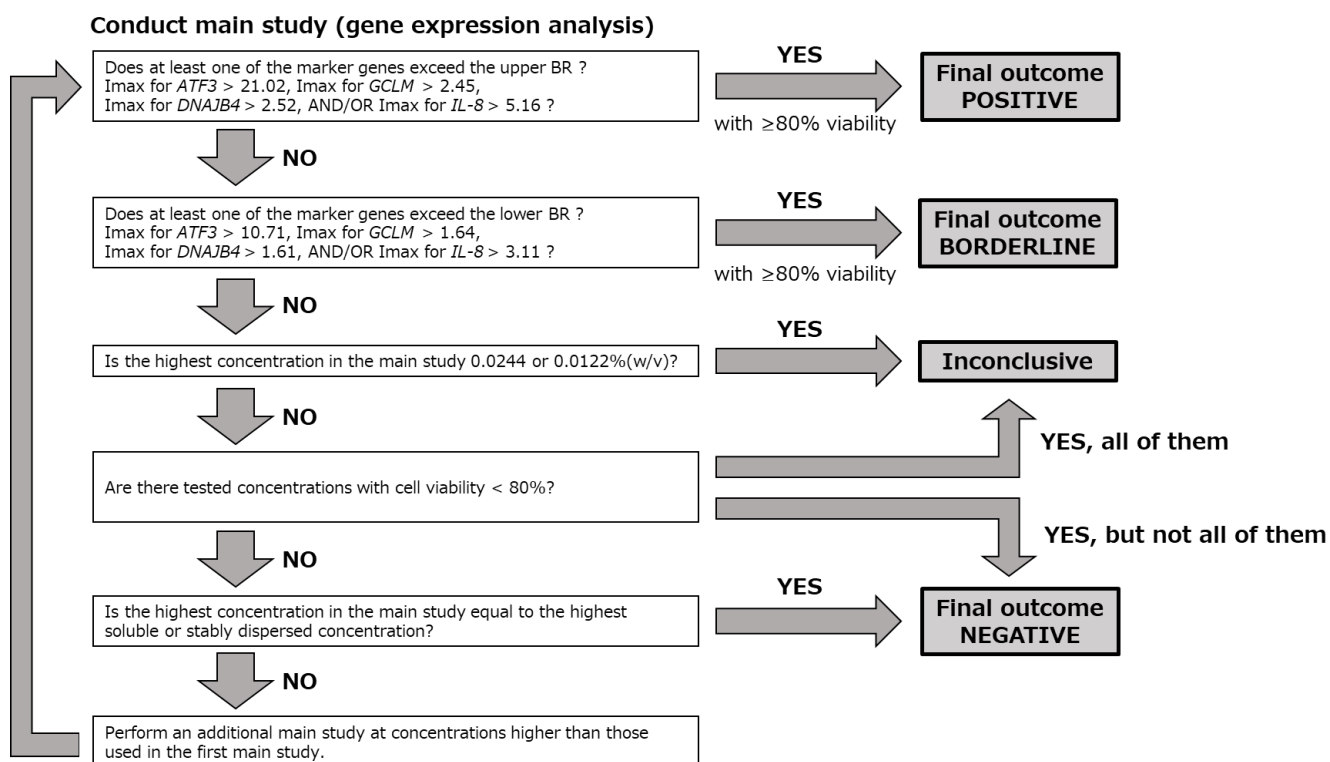
LuSens Prediction Model



EpiSensA Prediction Model

Figure A 1.1. Flow-chart of the EpiSensA prediction model, taking into account borderline ranges to conclude on borderline results within the 2o3 DA.

The original threshold for a positive classification for the four genes detected in the EpiSensA (see TG 442E) is ATF3 >15-fold, GCLM >2-fold, DNAJB4 > 2-fold, IL-8 > 4-fold induction with statistically derived borderline ranges of: $10.71 \leq \text{ATF3} \leq 21.02$; $1.64 \leq \text{GCLM} \leq 2.45$; $1.61 \leq \text{DNAJB4} \leq 2.52$; $3.11 \leq \text{IL-8} \leq 5.16$ fold induction.

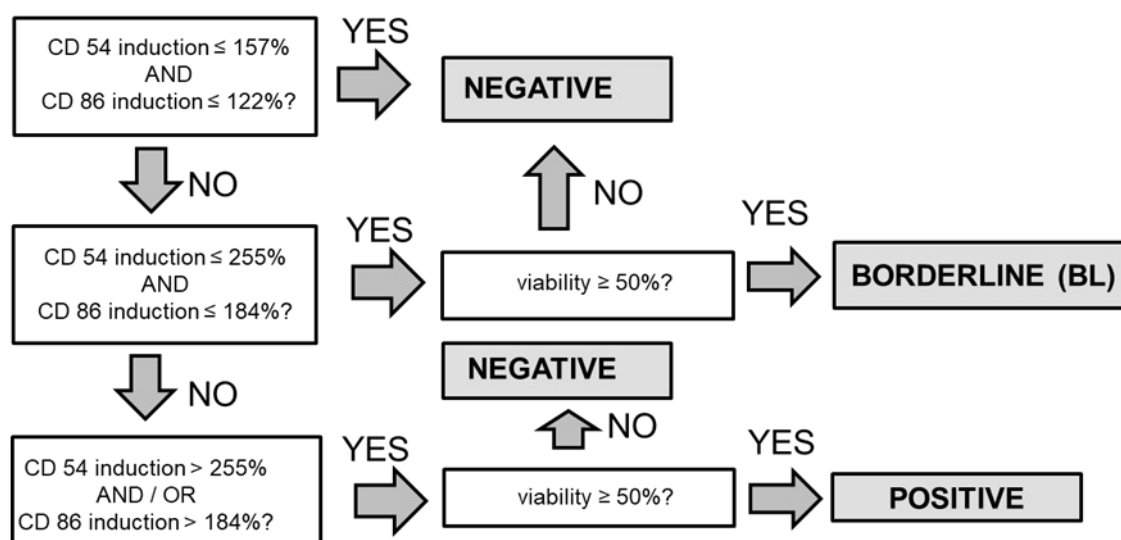


h-CLAT Prediction Model

Figure A 1.1. Flow-chart of the h-CLAT prediction model taking into account borderline ranges and multiple runs to conclude on borderline results within the 2o3 DA.

The original threshold for a positive classification is 150% induction of CD86 with a statistically derived borderline range around this threshold of 122 – 184% and 200% induction of CD54 with a statistically derived borderline range around this threshold of 157 – 255%.

Procedure for one full run:



Perform at least two independent runs.

- If two are **POSITIVE**, final outcome is **POSITIVE**
- If two are **NEGATIVE**, final outcome is **NEGATIVE**
- If two are **BL**, final outcome is **BL**

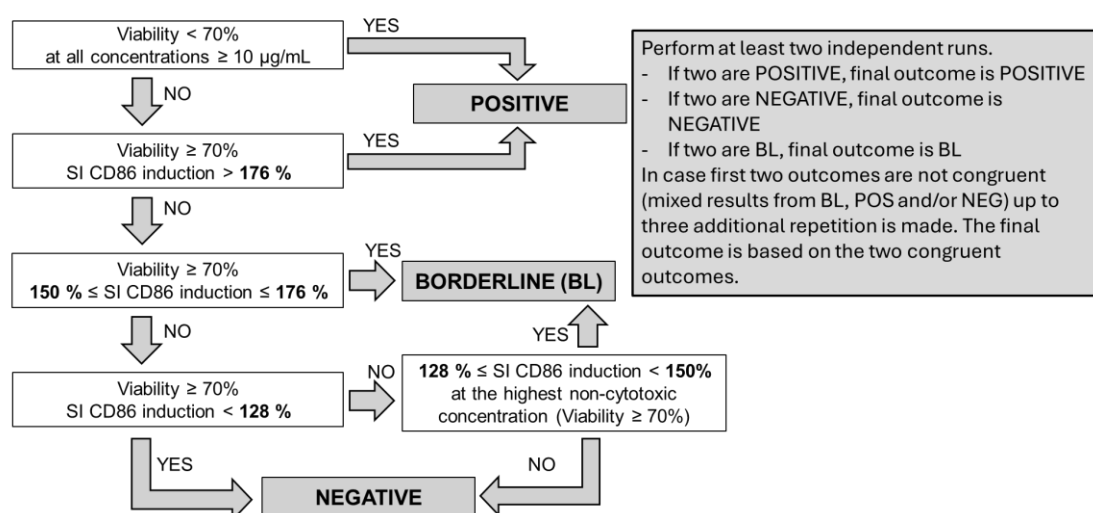
In case first two outcomes are not congruent (mixed results from BL, POS and/or NEG) a third repetition is made and final outcome is based on the two congruent outcomes.
[TWO BL] or [one BL, one NEG and one POS] ⇒ Final outcome BL

U-SENS Prediction Model

Figure A 1.1. Flow chart of the U-SENS™ prediction model taking into account borderline ranges and multiple runs to conclude on borderline results within the 2o3 DA.

The original threshold for a positive classification is 150% induction of the CD86 with a statistically derived borderline range around this threshold of 128% - 176%.

U-SENS™ Prediction Model Procedure for one full run:



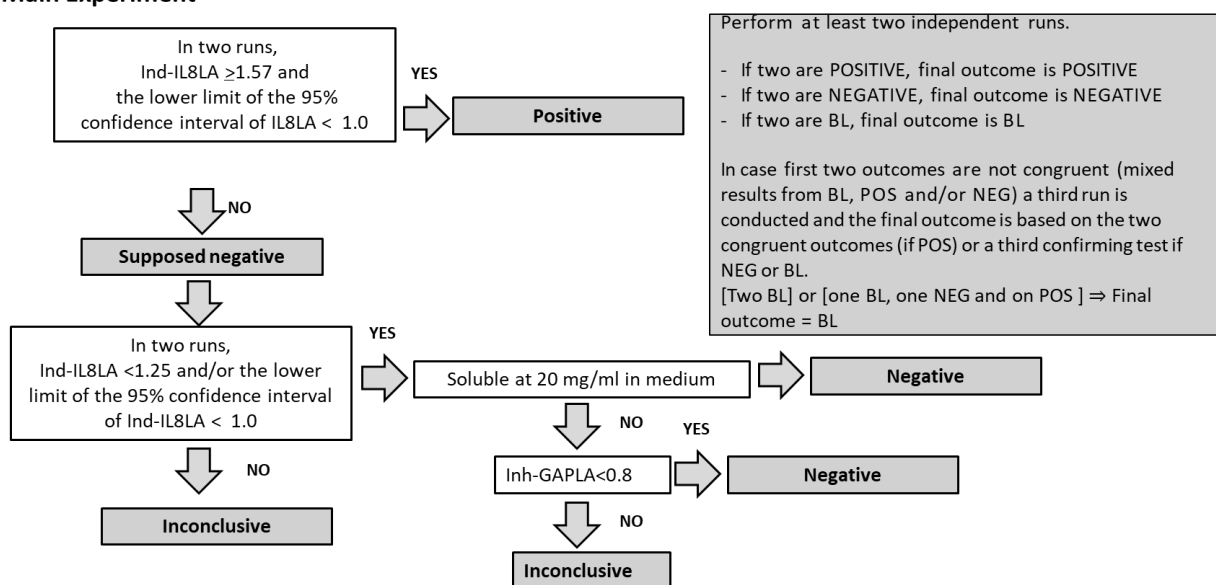
IL-8 Luc Prediction Model

Figure A 1.1. Flow chart of the IL-8 Luc assay prediction model taking into account borderline ranges and multiple runs to conclude on borderline results within the 2o3 DA.

The original threshold for a positive classification is 1.4-fold luciferase induction with a statistically derived borderline range around this threshold of 1.25 – 1.57.

IL-8 Prediction Model

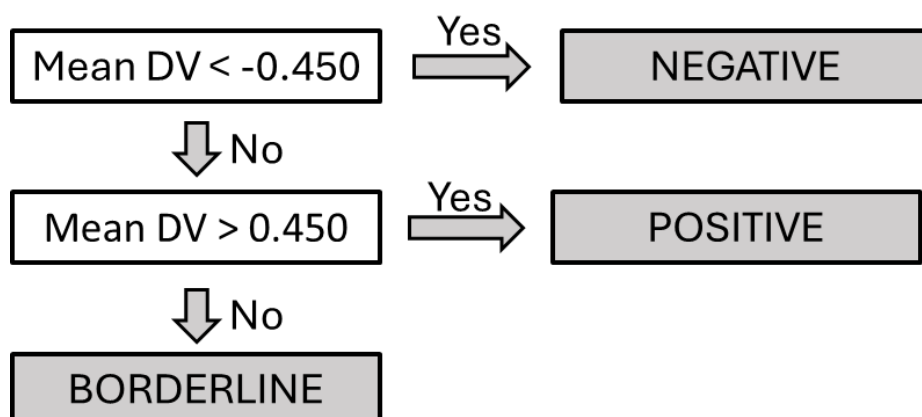
Conduct Main Experiment



GARDSkin Prediction Model

Figure A 1.1. Flow chart of the GARD[®]™ skin assay prediction model taking into account borderline ranges and multiple runs to conclude on borderline results within the 2o3 DA.

The original threshold for a positive classification is a mean DV > 0 with a statistically derived borderline range around this threshold of -0.450 – +0.450.

Procedure for one full run:

References

1. Kolle SN, Mathea M, Natsch A, Landsiedel R. (2021). Assessing Experimental Uncertainty in Defined Approaches: Borderline Ranges for *In Chemico* and *In Vitro* Skin Sensitization Methods Determined from Ring Trial Data, *Appl In Vitro Toxicol*, 7:3: 102-111.
2. *Correction to: Assessing Experimental Uncertainty in Defined Approaches: Borderline Ranges for In Chemico and In Vitro Skin Sensitization Methods Determined from Ring Trial Data*, by Kolle, et al. *Appl In Vitro Toxicol* 2021;7(3):102–111; doi: 10.1089/aivt.2021.0003. *Applied In Vitro Toxicology* 2024 10:1, 29-30.
3. Kasahara T, Yamamoto Y, Nakashima N, Imamura M, Mizumachi H, Suzuki S, Aiba S, Kimura Y, Ashikaga T, Kojima H, Ono A, Matsumoto K. (2025). Borderline Range Determined Using Data From Validation Study of Alternative Methods for Skin Sensitization: ADRA, IL-8 Luc Assay, and EpiSensA, *J Appl Toxicol*, 45(3):432-439.
4. Nardelli, L. (2025) Determination of U-SENS™ borderline range thresholds to address uncertainty for skin sensitization assessment in a regulatory context, *ALTEX-Alternatives to animal experimentation*. doi: 10.14573/altex.2411282.
5. OECD (2026), *Supporting document to the Guideline (GL) on Defined Approaches (DAs) for Skin Sensitisation*, Series on Testing and Assessment, No. 336, OECD Environment, Health and Safety, Paris. Available at: <https://www.oecd.org/en/topics/sub-issues/testing-of-chemicals/publications-on-testing-and-assessment-of-chemicals.html>

Annex 2. Assessing confidence in DASS ITS predictions

Introduction

116. As described in **Section 3.1**, the ITS defined approach (DA) are based on three information sources: an *in chemico* assay (DPRA, ADRA, TG 442C) (1), an *in vitro* assay (h-CLAT, U-SENS™, GARD®Skin, TG 442E) (2), and one *in silico* method (Derek Nexus or OECD Toolbox) prediction from the models listed in Appendices I-II of this guideline (referred to hereafter as *in silico*) (3-4). For each information source a score is given depending on the outcome of the individual assay and/or prediction, that is then summed to obtain the DA prediction (5).

Applicability domain of the individual information sources

In chemico/in vitro information source

117. A test chemical is considered to be within the *in chemico/in vitro* information source domain (i.e. applicable) if it can be tested according to the individual protocols, taking into account the technical and chemical type limitations of each assay (as defined in the respective TGs (442C and 442E (1, 2))). The *in chemico/in vitro* results are considered applicable, in case there are no technical or chemical space specific limitations and no reason why the results obtained from the assay cannot be considered.

In silico information source

118. The ITS DAs use *in silico* information sources that are based on chemical structures. These *in silico* sources rely on molecular representation of the chemicals: input usually by drawing the chemical structure, or by entering the Simplified Molecular-Input Line-Entry System (SMILES) or the IUPAC International Chemical Identifier (InChi). A test chemical is considered to be within the *in silico* information source domain if it is a single chemical (i.e. mixtures and UVCBs are outside the applicability domain) and a prediction can be made, taking into account the individual applicability domain of the *in silico* model (as defined in Appendix I-II). As a single chemical can be represented by several CAS or EC numbers (due to differences in composition, e.g. stereochemical differences, present as varied salt forms, present as the main component in a mixture), it is important to specify the exact structure. Resources such as the US EPA CompTox Chemicals Dashboard (<https://comptox.epa.gov/dashboard>) or NIH PubChem (<https://pubchem.ncbi.nlm.nih.gov/>) may be useful in mapping chemical names or structures to SMILES or

InChi format. Available guidance can be consulted regarding minimum purity level of substances used in *in silico* predictions based on molecular structure.¹⁷¹⁸

Confidence in ITS predictions

119. The applicability domain of the individual information sources used in the ITS DA are assessed and this determines whether the ITS predictions can be considered conclusive (i.e. high confidence) or inconclusive (i.e. low confidence) for hazard identification and/or potency.

How to apply the data interpretation procedure (DIP) for the ITS

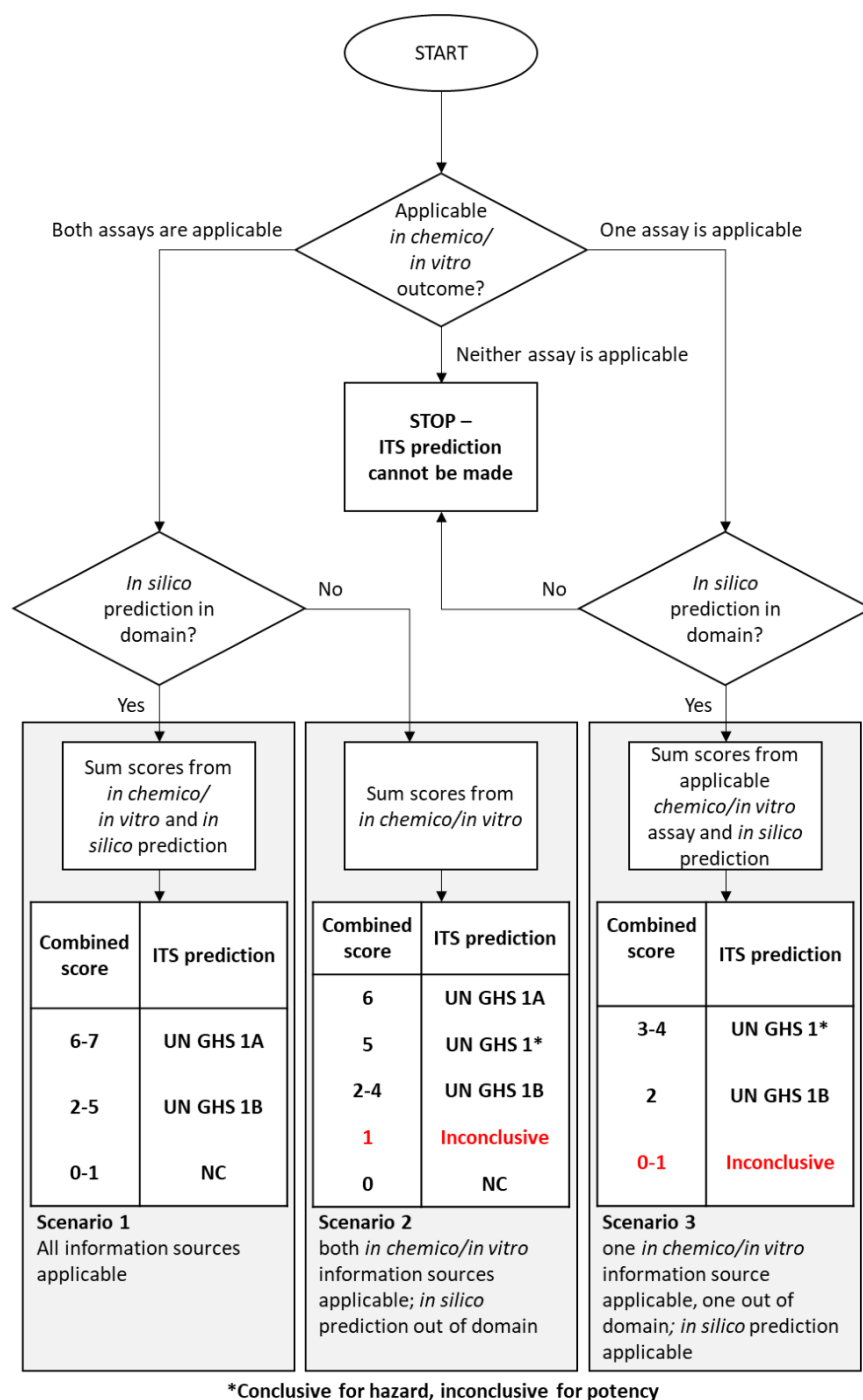
120. The ITS uses three information sources based on *in chemico* (KE1) (1) and *in vitro* (KE3) data (2), and an *in silico* prediction (3), with a DIP developed with EG DASS input. Where all three information sources are applicable, a conclusive ITS prediction can be made. In some cases, a conclusive ITS prediction can be made, if there are only two information sources with applicable results (**Figure A2.1**).

¹⁷ OECD (2017), Guidance on Grouping of Chemicals, Second Edition, OECD Series on Testing and Assessment, No. 194, OECD Publishing, Paris, <https://doi.org/10.1787/9789264274679-en>.

¹⁸ ECHA (2008) CHAPTER R.6 – QSARS AND GROUPING OF CHEMICALS *in* Guidance on

Information Requirements and Chemical Safety Assessment. European Chemicals Agency [[Guidance on Information Requirements and Chemical Safety Assessment - ECHA \(europa.eu\)](https://echa.europa.eu/guidance-on-information-requirements-and-chemical-safety-assessment)]

Figure A 2.1. Workflow for data interpretation procedure for the ITS



Note: The ITS defined approach (DA) are based on three information sources: an *in chemico* assay (DPRA, ADRA, TG 442C) (1), an *in vitro* assay (h-CLAT, U-SENS™, GARD@Skin, TG 442E) (2), and one *in silico* method (Derek Nexus or OECD Toolbox prediction from the models listed in Appendices I-II of this guideline) (3-4). For each information source a score is given depending on the outcome of the individual assay and/or prediction, that is then summed to obtain the DA prediction. Scoring information can be found in Table 3.1.

121. Depending on the applicability of the individual information sources, three different scenarios for the ITS DA are possible (see **Appendices I-II, Annex 2 Figure A 2.1. and Annex 2 Table A 2.1.**). In Scenario 1, all three information sources are applicable. In Scenarios 2 and 3, only two information sources are applicable. Details are provided below:

122. Scenario 1: all of the information sources i.e. *in chemico/in vitro* outcomes are applicable and can be considered (as prescribed in each individual assay) and the *in silico* prediction is in domain. The obtained ITS DA prediction is conclusive and of high confidence

123. Scenario 2: *in silico* prediction out of domain, however *in chemico/in vitro* methods are in domain and provide conclusive predictions (i.e. *in chemico/in vitro* methods are applicable).

- Combined DA score of 0, 2, 3, 4 or 6, *in silico* prediction out of *in silico* domain: DA conclusion is possible based on the two *in chemico/in vitro* outcomes. Conclusive prediction as the *in silico* prediction would not lead to a different DA prediction.
- Combined DA score of 5, *in silico* prediction out of *in silico* domain: DA conclusion possible for hazard identification (conclusive positive DA prediction for hazard identification). DA conclusion not possible for potency (inconclusive DA prediction for potency).
- Combined DA score of 1, *in silico* prediction out of *in silico* domain: DA conclusion not possible. Inconclusive DA prediction for hazard identification and potency.

124. Scenario 3: one *in chemico/in vitro* method out of domain or the result of that method cannot be considered (inapplicable):

- Combined DA score of 2 based on one *in chemico/in vitro* and *in silico* prediction: DA conclusion possible. Conclusive DA prediction as UN GHS 1B, as the outcome of the other *in chemico/in vitro* method would not lead to a different DA prediction.
- Combined DA score of 3 or 4, based on one *in chemico/in vitro* and *in silico* prediction: DA conclusion possible for hazard identification (conclusive positive DA prediction for hazard identification). DA conclusion not possible for potency (inconclusive DA prediction for potency).
- Combined DA score of 0 or 1, one *in chemico/in vitro* and *in silico* prediction: DA conclusion not possible. Inconclusive prediction for hazard identification and potency.

Table A 2.1. Applicability domain and confidence of the ITS.

Scenario	Combined score ¹⁹	ITS prediction	Confidence	DA prediction including confidence considerations
1	0-1	NC	High	Conclusive prediction Not Classified (NC).
	2-5	UN GHS 1B	High	Conclusive prediction UN GHS 1B.
	6-7	UN GHS 1A	High	Conclusive prediction UN GHS 1A.
2	0	NC	High	Conclusive prediction NC.
	1	Inconclusive	Low	Inconclusive prediction whether positive or negative.
	2-4	UN GHS 1B	High	Conclusive prediction UN GHS 1B.
	5	UN GHS 1	High	Conclusive positive prediction for hazard identification.
			Low	Inconclusive prediction for potency.
	6	UN GHS 1A	High	Conclusive prediction UN GHS 1A.
3	0-1	Inconclusive	Low	Inconclusive prediction whether positive or negative.
	2	UN GHS 1B	High	Conclusive prediction UN GHS 1B.
	3-4	UN GHS 1	High	Conclusive positive prediction for hazard identification.
			Low	Inconclusive prediction for potency.

¹⁹ Total scores calculated only from information sources that are applicable/in domain.

References

1. OECD (2026), *Test No. 442C: In Chemico Skin Sensitisation: Assays addressing the Adverse Outcome Pathway key event on covalent binding to proteins*, OECD Guidelines for the Testing of Chemicals, Section 4, OECD Publishing, Paris, <https://doi.org/10.1787/9789264229709-en>.
2. OECD (2026). *Test No. 442E: In Vitro Skin Sensitisation: In Vitro Skin Sensitisation assays addressing the Key Event on activation of dendritic cells on the Adverse Outcome Pathway for Skin Sensitisation*, OECD Guidelines for the Testing of Chemicals, Section 4, OECD Publishing, Paris, <https://doi.org/10.1787/9789264264359-en>.
3. Lhasa Limited (2020), Derek Nexus v6.1.0, Lhasa Limited. (<https://www.lhasalimited.org/products/derek-nexus.htm>)
4. Yordanova et al., (2019). Automated and standardized workflows in the OECD QSAR Toolbox. *Computational Toxicology*, 10, 89–104. <https://doi.org/10.1016/j.comtox.2019.01.006>
5. Takenouchi O, Fukui S, Okamoto K, Kurotani S, Imai N, Fujishiro M, Kyotani D, Kato Y, Kasahara T, Fujita M, Toyoda A, Sekiya D, Watanabe S, Seto H, Hirota M, Ashikaga T, Miyazawa M. (2015). Test battery with the human cell line activation test, direct peptide reactivity assay and DEREK based on a 139 chemical data set for predicting skin sensitizing potential and potency of chemicals. *J Appl Toxicol*, 35:1318-32

Appendix I: Description, applicability domain, and protocol for Derek Nexus predictions

Description of Derek Nexus

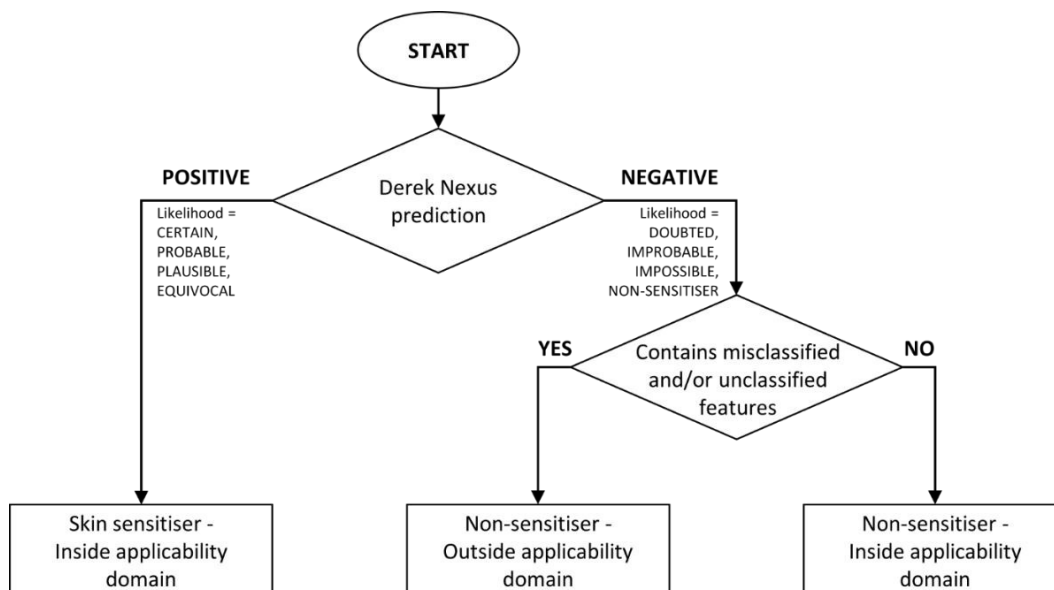
125. Derek is an expert, knowledge-based software tool comprising alerts on several toxicity endpoints, including skin sensitisation. Derek fires alerts based on structural features, *i.e.* whether a hapten has potential for electrophilic binding to skin proteins either directly or following metabolism/auto-oxidation (1). Additional information on this model can be found in the QSAR Model Reporting Format (QMRF) (2).

Applicability domain of Derek Nexus

126. To each skin sensitisation alert, a likelihood level is associated. Chemicals firing an alert with a likelihood of certain, probable, plausible, or equivocal are considered to be positive. Chemicals with a negative prediction of 'non-sensitiser with no misclassified or unclassified features' are considered to be negative.

127. Skin sensitisation predictions from Derek Nexus v6.1.0 are used in the ITS DA. All positive predictions (likelihood = certain, probable, plausible or equivocal) are considered to be inside the applicability domain. Negative predictions (likelihood = doubted, improbable, impossible or non-sensitiser) are also considered to be in the applicability domain unless they contain misclassified and/or unclassified features. A prediction of non-sensitiser with misclassified features indicates the presence of a fragment that has been observed exclusively in known sensitisers which Derek fails to alert for. A prediction of non-sensitiser with unclassified features indicates the presence of a fragment that has not been observed in publicly available data (although Derek may have seen this in proprietary data) (3). Usually, expert review is recommended for predictions containing these features but as a fixed data interpretation procedure, required in a DA, does not permit expert review these are best considered as out of domain for use in the ITS DA (**Figure 8**).

Figure 8. Applicability domain for Derek Nexus skin sensitisation predictions used in the ITS DA



Protocol for generating predictions for skin sensitisation hazard using Derek Nexus v.6.1.0 with Derek KB 2020 1.0

128. The following protocol should be used to generate predictions for skin sensitisation hazard using Derek Nexus v.6.1.0 with Derek Knowledge Base (KB) 2020 1.0 to be used as the *in silico* information source for the ITSv1 defined approach.

Single chemical

1. Open Nexus
2. Input structure using one of the following options:
 - a. Input structure manually by drawing on the canvas
 - b. Go to File>Open Structure(s) to input a single structure from a file (.mol, .sdf, .smi, .csv, .cdx (file list not exhaustive))
 - c. Go to File>Type Chemistry to enter or paste SMILES, InChi or MOL file
 - d. Go to File>New Structure to input structure by drawing a structure
3. Set up prediction
 - a. Go to Prediction>Derek Prediction>Derek Prediction Setup
4. Apply processing constraints
 - a. Knowledge Bases
 - i. For Nexus v6.1.0, ensure Derek KB 2020 1.0 is selected
 - b. Perception
 - i. Ensure 'Perceive tautomers' and Perceive mixtures' are selected

- ii. Ensure 'Match alerts without rules' is unselected
 - c. Species
 - i. Select 'mammal'
 - d. Endpoints
 - i. Click 'Deselect all' then expand 'Skin sensitisation (ALL)' to view 'Photoallergenicity' and 'Skin sensitisation'. Select 'Skin sensitisation'
 - e. Structure properties
 - i. Ensure the 'Overwrite' box(es) for logP, logKp, and average molecular mass are unselected to use the values calculated by Derek Nexus, otherwise, check the 'Overwrite' box(es) to input own values.
- 5. Generate prediction
 - a. Click 'Start Prediction'
 - b. If an alert is fired: Knowledge base, endpoint, species, reasoning level, alert fired, EC3 prediction (if applicable), and example matched (if applicable) are shown in the prediction navigator.
 - i. Click the likelihood (certain, probable, plausible, equivocal) to view the reasoning rules leading to the likelihood level.
 - ii. Click the Alert in the prediction navigator to view alert match(es), description image, comments, validation comments, endpoint, references, patterns, and examples associated with the alert.
 - c. If no alert is fired, a negative prediction is generated: Knowledge base, endpoint, species and negative prediction reasoning (non-sensitiser) and negative prediction overview (absence or presence of misclassified and/or unclassified features) are shown in the prediction navigator.
 - i. Click the negative prediction overview ('No misclassified or unclassified features', 'Contains misclassified/unclassified features') to view information about the negative prediction. Similar nearest neighbours are available to view for misclassified features.
 - d. Use the Derek likelihood to classify each compound as positive or negative (alert fired with certain, probable, plausible, or equivocal is classified as positive, alert fired with doubted, improbable, impossible, or a negative prediction of non-sensitiser with no misclassified or unclassified features is classified as negative).
 - i. Negative predictions of non-sensitiser with misclassified and/or unclassified features are of lower confidence and are not used in ITSv1.
 - ii. In cases where more than one alert is fired or structures in a mixture generate different likelihoods, the most conservative classification is applied (positive > negative).
 - iii. A positive outcome from Derek is scored as 1 in the ITSv1 and a negative outcome is scored as 0.

Multiple chemicals

1. Open Nexus
2. Input structures
 - a. Go to File>Open Structure(s) to input a file containing multiple structures (.mol, .sdf, .smi, .csv, .cdx (file list not exhaustive))
 - b. Select the fields from the file which will be mapped to structure properties used during the prediction (Name, Average Molecular Mass, LogP, LogKp). If left unchanged then the values set by Derek will be used.
3. Set up batch prediction
 - a. Go to Prediction>Derek Prediction>Derek Batch Setup
4. Apply processing constraints
 - a. Knowledge Bases
 - i. For Nexus v6.1.0, ensure Derek KB 2020 1.0 is selected
 - b. Perception
 - i. Ensure 'Perceive tautomers' and 'Perceive mixtures' are selected
 - ii. Ensure 'Match alerts without rules' is unselected
 - c. Species
 - i. Select 'mammal'
 - d. Endpoints
 - i. Click 'Deselect all' then expand 'Skin sensitisation (ALL)' to view 'Photoallergenicity' and 'Skin sensitisation'. Select 'Skin sensitisation'
 - e. Report configuration
 - i. Directory - Leave as default directory or map to preferred location.
 - ii. Pick type - Select report for batch (left side icon)
 - iii. Pick format - Select desired file type (e.g. Excel)
 - iv. Pick design - Select desired design (e.g. Tabular Report)
 - v. Filename - input desired filename
 - f. Report display options
 - i. Ensure 'Show predictions of at least impossible' is selected
 - ii. Select 'Show Negative Predictions'
 - iii. Select 'Filter All Nearest Neighbours by Misclassified Features'
 - iv. Select 'Show Open Likelihood'
 - v. Select 'Show Rapid Prototypes'
5. Generate batch prediction
 - a. Click 'Start Batch Prediction'
 - i. Once the batch prediction is finished, select the 'Open Report Directory' when prompted
 - b. Use the Derek likelihood to classify each compound as positive or negative (alert fired with certain, probable, plausible, or equivocal is classified as positive, alert fired with doubted, improbable, impossible, or a negative prediction of non-sensitiser with no misclassified or unclassified features is classified as negative).
 - i. Negative predictions of non-sensitiser with misclassified and/or unclassified features are of lower confidence and are not used in ITSv1.

- ii. In cases where more than one alert is fired or structures in a mixture generate different likelihoods, the most conservative classification is applied (positive > negative).
- c. A positive outcome from Derek is scored as 1 in the ITSv1 and a negative outcome is scored as 0.

References

1. Lhasa Limited (2020), Derek Nexus v6.1.0, Lhasa Limited. (<https://www.lhasalimited.org/products/derek-nexus.htm>)
2. OECD (2024) Webpage for Software to be used with Test Guidelines: <https://www.oecd.org/en/topics/sub-issues/testing-of-chemicals/software-test-guidelines.html>
3. Chilton, M. L., Macmillan, D. S., Steger-Hartmann, T., Hillegass, J., Bellion, P., Vuorinen, A., Etter, S., Smith, B. P. C., White, A., Sterchele, P., De Smedt, A., Glogovac, M., Glowienke, S., O'Brien, D., & Parakhia, R. (2018). Making reliable negative predictions of human skin sensitisation using an in silico fragmentation approach. *Regulatory Toxicology and Pharmacology*, 95, 227–235. <https://doi.org/10.1016/j.yrtph.2018.03.015>

Appendix II: Description, applicability domain, and protocol for OECD QSAR Toolbox predictions

Description of OECD QSAR Toolbox

129. The OECD QSAR Toolbox is a freely available software application that supports reproducible and transparent chemical hazard assessment. It offers functionalities for retrieving experimental data, simulating metabolism and profiling properties of chemicals. These information and tools can be used to find structurally and mechanistically defined analogues and chemical categories, which can serve as sources for read-across and trend analysis for data gap filling (1)]. Additional information on this model can be found in the QSAR Model Reporting Format (QMRF) (2).

Applicability domain of OECD QSAR Toolbox

Technical aspects

130. The Toolbox prediction used by the DA ITS is calculated using the DASS automated workflow (DASS AW) included in OECD QSAR Toolbox v.4.5. The workflow also includes the automatic calculation of the applicability domain.

Calculation of the in silico domain of Toolbox

131. Applicability domain of the QSAR Toolbox Skin sensitisation predictions for use in the ITS defined approach automated workflow (DASS AW) is defined by based on the training set substances of the same automated workflow. The training set (TS) consists of 2268 substances having LLNA and/or GPMT skin sensitisation experimental data²⁰ (the full list of substances can be consulted in the QSAR Toolbox). The TS substances are part of the following OECD QSAR Toolbox databases:

1. Skin sensitisation;
2. REACH Skin sensitisation (normalized) databases.

132. Based on the correctly predicted training set substances, three layers of applicability domain are automatically calculated by the Toolbox: 1) parametric; 2) structural and 3) mechanistic layers. Depending on the Toolbox prediction approach (read-across or profiling predictions) and prediction outcomes (positive or negative), one or more of these layers are taken into account to establish the overall Toolbox domain of the specific prediction.

²⁰ In case of multiple data points for one substance, the most conservative scenario is taken into account.

133. The applicability domain layers considered for different types of Toolbox predictions are summarised in the table here:

Toolbox DASS AW outcome		Applicability domain layer		
		Structural	Parametric	Mechanistic
Positive	Read-across	Not considered	Not considered	Considered
	Profiling	Not considered	Not considered	Met by definition
Negative	Read-across	Not considered	Not considered	Considered
	Profiling	Considered	Considered	Met by definition

134. Explanation and rationale for the use of different domain layers:

1. Positive predictions (both by read-across and profiling): the presence of an alert (which is the requirement for positive Toolbox prediction to be considered within in the mechanistic domain) is sufficient to consider the prediction to be within the Toolbox domain. Substances triggering an alert are considered as in domain because they contain the toxicophore that has been observed experimentally in skin sensitisers. No further checks are needed in this context to consider the prediction within the Toolbox *in silico* domain.
2. Negative predictions by read-across: the structural and parametric domains are not taken into account because the Toolbox has already ensured some level of similarity with other substances in its training set that met the requirements to be selected as suitable analogues for read-across (these requirements are explained in detail in the DASS AW description).
3. Negative prediction by profiling predictions: all domain layers are taken into account to ensure the highest possible reliability level for the Toolbox prediction. Stricter requirements are needed mainly for two reasons: 1. lack of alerts is not equal to proof of lack of sensitisation potential and 2. to apply a cautious approach since acceptance of negative predictions may lower the human health protection level risk in case of a false negative predictions.

Calculation of applicability domain layers

1. Parametric layer

Four physicochemical parameters of the substances are taken into consideration: log Kow, molecular weight, vapour pressure and water solubility²¹. The ranges of variation for the selected parameters are defined based on the training set substances that are correctly predicted by the DASS AW.

A substance is considered within the parametric domain of the DASS AW if its physicochemical parameter values as calculated by the QSAR Toolbox fall into the ranges of variation given in the table below. It is noted that the ranges include parametric values calculated using EPISuite models implemented in Toolbox that in some cases are wider than that covered by existing test methods.

Physicochemical parameter	Calculated Parameter range
Log Kow	-9.66 ÷ 18.6

²¹ QSAR Toolbox is used for the calculation of the physico-chemical properties.

Molecular weight	16 Da ÷ 2290 Da
Vapour pressure*	0 Pa ÷ 3.45×10^7 Pa
Water solubility	2.48×10^{-15} mg/L ÷ 1.00×10^6 mg/L

*EPIWIN Vapor Pressure (Antoine method) is used for calculation

2. Structural layer

The structural layer is defined based on the atom centred fragments (ACF) derived from the structural characteristics of the TS substances that are correctly predicted²² by the DASS AW.

The ACF are defined according to the following Toolbox default values for ACF:

- Any atom distance = 1
- Heteroatom distance = 1
- Extract C (sp³) fragments = YES
- Include whole aromatic rings = NO

For each substance, the following values are calculated:

- % Correct fragments: percentage of ACF occurring in correctly predicted structures in the training set
- % incorrect fragments: percentage of ACF occurring in incorrectly predicted structures in the training set
- % unknown fragments: percentage of ACF not occurring in the training set.

A substance is considered within the structural domain of the DASS AW if 100% of its ACF belong to the correct fragments.

3. Mechanistic layer

The predicted capability of a substance to interact with the skin proteins without and after (a)biotic activation is taken into consideration. The Toolbox endpoint-specific profiler *Protein binding for skin sensitization by OASIS* and two metabolic simulators – *Autoxidation simulator* and *Skin metabolism simulator* are used to predict such interaction.

A positive prediction is considered within the mechanistic domain if the substance triggers alerts from the profiler “*Protein binding for skin sensitization by OASIS*”—without or after (a)biotic activation.

A negative prediction is considered within the mechanistic domain if the substance does not trigger alerts from the profiler “*Protein binding for skin sensitization by OASIS*” without or after (a)biotic activation.

Note that predictions obtained by profiling results will meet the mechanistic layer requirements by definition because positive Toolbox predictions by profiler are triggered exactly by the presence of alert. If the test

²² All ACF that are extracted from the correctly predicted TS test chemicals “good space”. The “bad space” is formed from the ACF present in the incorrectly predicted test chemicals. The default QSAR Toolbox settings for ACF are used. Supplementary file with the ACF forming the good and the bad space are available.

chemical cannot be tested or the outcome/prediction cannot be considered in at least two of the information sources (*in chemico/in vitro and/or in silico*) then the DA cannot be applied.

Protocol for generating predictions for skin sensitisation hazard using DASS AW in Toolbox 4.5.

135. The following protocol should be used to generate predictions for skin sensitisation hazard using OECD QSAR Toolbox v.4.5 with the automated workflow for defined approaches for skin sensitisation (DASS AW) to be used as the *in silico* information source for the ITSv2 defined approach.

Step 1: Input the chemical in the “Input module”. SMILES is the preferred way to input the structure. (If other identifiers such as the CAS number are used as input, the Toolbox will assign the SMILES based on its internal database. In this case, the user needs to make sure that Toolbox identifies and consequently uses for the prediction the correct structure.)

Step 2: Go to the “Data gap filling module” and click on “Automated” button. Select “EC3 from LLNA or Skin sensitization from GPMT assays for defined approaches” and click OK. The scheme with the implemented logic will be shown.

Step 3: Click the Run button -  or press F5 key of the keyboard and confirm with “Yes”. The workflow will run automatically.

Step 4: If a substance is predicted “positive” or “negative” as a result of read-across, the prediction will appear on the data matrix with “R” in front of the result (e.g. “R: Negative”). If a substance is predicted “positive” or “negative” as a result of profiling, then the result will appear next to the name of the customized profiler “Skin sensitization for DASS”.

Step 5: Affiliation of the substance to the domain of the automated workflow for DASS will be automatically determined and presented.

References

1. OECD QSAR Toolbox, <https://qsartoolbox.org/>, accessed: 20 February 2025.
2. OECD (2024) Webpage for Software to be used with Test Guidelines: <https://www.oecd.org/en/topics/sub-issues/testing-of-chemicals/software-test-guidelines.html>

Appendix III: Protocol for Generating SARA-ICE Predictions

The following protocol should be used to generate predictions for point of departure for skin sensitisation risk assessment using SARA-ICE DA v.1.0.

1. Establishing SARA-ICE Inputs

1. Click 'Download Template'
*An example template is available if needed. Click 'Download Example'.
A key is provided on the final tab.*
2. Select 'Substances' tab. Add distinct chemical substances (1 or more).
 - a. Input name for the chemical(s) in column A 'Substance Name'
 - b. *Optional:* Add an abbreviated/shorter substance name that will appear in tables, figures etc. in column B 'Abbreviated Substance Name'.
If this is not used the tool will automatically revert to the substance name in column A.
 - c. Input CAS number (or other unique identifier) in column C
Whichever unique identifier is used here will also be used within the data tabs to ensure data for this chemical is linked across data types
 - d. Input Molecular weight (MW) in g/mol, in column D, if using kDPRA and/or KeratinoSens data inputs. This allows automatic conversion from molar to mass concentrations.
3. Populate data tabs

Although the model can technically be run with no chemical specific information (in which case the resulting ED₀₁ estimates will be taken directly from the prior distribution), any inputted chemical specific data will inform the ED₀₁ distribution. A minimum of 2 test methods per compound, covering minimum of two key events, should be entered into the data template as per the guideline to ensure adequate uncertainty reduction in the ED₀₁ distribution. Reference methods which can be used as SARA-ICE DA inputs include: the Direct Peptide Reactivity Assay (DPRA; OECD TG 442C), kinetic DPRA (kDPRA; OECD TG 442C), KeratinoSens (OECD TG 442D), the human cell line activation test (h-CLAT; OECD TG 442E), and U-SENS (OECD TG 442E), as well as in vivo methods such as the local lymph node assay (LLNA; OECD TG 429, OECD TG 442A, OECD TG 442B) and HPPT test methods human repeat insult patch tests and human maximization tests. No new in vivo data should be produced for use in SARA-ICE DA.

- a. HPPT
 - i. *Optional:* Input 'Substance Name' in column A
 - ii. Input CAS number (or other chosen unique identifier) in column B
Whichever unique identifier is used here will also be used within the data tabs to ensure data for this chemical is linked across data types
 - iii. Input dose per skin area (µg/cm²) in column C
 - iv. Input number of test subjects in column D
 - v. Input number of test subjects with positive sensitisation responses (post elicitation phase of the HPPT study) in column E
 - vi. *Optional:* Input data source reference in column F

Note, this is not used within the model but will be included in output reports

- b. LLNA
 - i. *Optional:* Input 'Substance Name' in column A
 - ii. Input CAS number (or other chosen unique identifier) in column B
Whichever unique identifier is used here will also be used within the data tabs to ensure data for this chemical is linked across data types
 - iii. Input the (estimated) EC3 (%) in column C
Negative studies should be entered as >X where X is the maximum concentration tested. Studies where the SI > 3 at the lowest reported/tested concentration should be entered as <X where X is the lowest concentration reported/tested.
 - iv. *Optional:* Input data source reference in column D
Note, this is not used within the model but will be included in output reports
- c. DPRA
 - i. *Optional:* Input 'Substance Name' in column A
 - ii. Input CAS number (or other chosen unique identifier) in column B
Whichever unique identifier is used here will also be used within the data tabs to ensure data for this chemical is linked across data types
 - iii. Input percentage depletion of the cysteine peptide in column C
 - iv. Input percentage depletion of the lysine peptide in column D
 - v. *Optional:* Input data source reference in column E
Note, this is not used within the model but will be included in output reports
- d. KDPRA
 - i. *Optional:* Input 'Substance Name' in column A
 - ii. Input CAS number (or other chosen unique identifier) in column B
Whichever unique identifier is used here will also be used within the data tabs to ensure data for this chemical is linked across data types
 - iii. Input log Kmax value in column C
When reactivity rate is too low to measure, input as <X where X is the lowest possible measurable reactivity rate. Typically $X = -3.5 \text{ M}^{-1} \text{ s}^{-1}$
 - iv. *Optional:* Input data source reference in column D
Note, this is not used within the model but will be included in output reports
- e. KeratinoSens™
 - i. *Optional:* Input 'Substance Name' in column A
 - ii. Input CAS number (or other chosen unique identifier) in column B
Whichever unique identifier is used here will also be used within the data tabs to ensure data for this chemical is linked across data types
 - iii. Input EC1.5 (µM) (concentration resulting in a 1.5-fold increase in luciferase activity) in column C
 - iv. Input IC50 (µM) (concentration resulting in 50% cell death in the KeratinoSens study) in column D
 - v. *Optional:* Input data source reference in column E
Note, this is not used within the model but will be included in output reports
- f. h-CLAT
 - i. *Optional:* Input 'Substance Name' in column A
 - ii. Input CAS number (or other chosen unique identifier) in column B
Whichever unique identifier is used here will also be used within the data tabs to ensure data for this chemical is linked across data types
 - iii. Input CD54 EC200 (µg/ml) in column C
Concentration resulting in a 200% increase in expression of the CD54 biomarker. Input can be entered as >X where X is the maximum concentration tested if not concentration-dependent response is observed in the assay. If a biomarker shows a response at the minimum tested concentration, input result as <X where X is the minimum concentration tested.
 - iv. Input CD86 EC150 (µg/ml) in column D

Concentration resulting in a 150% increase in expression of the CD86 biomarker. Input can be entered as >X where X is the maximum concentration tested if not concentration-dependent response is observed in the assay. If a biomarker shows a response at the minimum tested concentration, input result as <X where X is the minimum concentration tested.

- v. Input CV75 (µg/ml) in column E
Concentration at which there is 75% cell viability.
- vi. *Optional:* Input data source reference in column F
Note, this is not used within the model but will be included in output reports
- g. USENS™
 - i. *Optional:* Input 'Substance Name' in column A
 - ii. Input CAS number (or other chosen unique identifier) in column B
Whichever unique identifier is used here will also be used within the data tabs to ensure data for this chemical is linked across data types
 - iii. Input EC150 (µg/ml) in column C
Concentration resulting in a 150% increase in expression of the CD86 biomarker. Input can be entered as >X where X is the maximum concentration tested if not concentration-dependent response is observed in the assay. If a biomarker shows a response at the minimum tested concentration, input result as <X where X is the minimum concentration tested.
 - iv. Input CV70 (µg/ml) in column D
Concentration at which there is 70% cell viability
 - v. *Optional:* Input data source reference in column E
Note, this is not used within the model but will be included in output reports
- 4. Save completed template file

2. Running the SARA-ICE Model

1. Click 'Choose input' on the SARA-ICE DA user interface
2. Select saved completed template file
3. Click 'Run Analysis'

Running the model involves drawing samples from the posterior distribution using the No-U-Turn sampler implemented within the probabilistic programming language Stan. By default, a sample of 10,000 draws of the ED_{01} is returned to the user. Summary statistics such as quantile estimates of the distribution are calculated empirically on this sample.

The No-U-Turn sampler produces a Markov Chain whose stationary distribution is the posterior distribution of the model. A successful model run is achieved when the Markov Chain converges to the target distribution. The Stan software provides several measures of convergence to assess whether convergence may be assumed.

3. SARA-ICE Model Outputs

1. The 5th, 50th and 95th percentiles of the ED_{01} (µg cm⁻²) are provided in tabular format and the prediction interval is presented in a graphical format within the user interface.
These data are also available within a downloadable template format (.xlsx or .csv), click 'Download Analysis' for the appropriate file format. The downloadable template also provides the ED_{01} at all percentiles ranging from 0.5th to 95.5th (in 0.5 increments). This template also provides the full set of draws from which summary statistics are computed.
2. The SARA-ICE PoD (computed as the geometric mean of the ED_{01} distribution conditional on it being less than 60,000 µg cm⁻²) is provided in a tabular format within the user interface.

A conservative PoD (calculated at the 5th percentile of the ED01 conditional on it being less than 60,000 µg cm⁻² is also provided and these data are also available within a downloadable template format (.xlsx or .csv), click 'Download Analysis' for the appropriate file format.

3. The SARA-ICE probability of binary (S/NC) and GHS subcategory (1A, 1B, NC) are given in a tabular format within the user interface.

Note, this is a capability of the SARA-ICE model shared through the user interface but does not form part of the SARA-ICE DA within OECD TG 497. These data are also available within a downloadable template format (.xlsx or .csv), click 'Download Analysis' for the appropriate file format.

4. The SARA-ICE GHS category call is provided in a tabular format within the user interface. The classification is dependent on the confidence in the prediction, e.g., a probability threshold. The probability threshold for a high-confidence prediction must be higher than the prior probability of 0.67. Sliders within the user interface are provided to the user in order to set the probability threshold as per their requirements.

Note, this is a capability of the SARA-ICE model shared through the user interface but does not form part of the SARA-ICE DA within OECD TG 497. A full sensitivity analysis on the impact of varying the probability threshold can be found in supporting documentation. These data are also available within a downloadable template format (.xlsx or .csv), click 'Download Analysis' for the appropriate file format.